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ANALYTICAL MODELS FOR FINANCIAL MODELING AND RISK MANAGEMENT

Risk minimization in multi-factor portfolios: What is the best strategy?

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Abstract Exposures to risk factors, as opposed to individual securities or bonds, can lead to an ex-ante improved risk management and a more transparent and cheaper way of developing active asset allocation strategies. This paper provides an extensive analysis of eight state-of-the-art risk-minimization schemes and compares risk factor performance in a conditional performance analysis, contrasting good and bad states of the economy. The investment universe spans a total of 25 risk factors, including size, momentum, value, high profitability and low investments, from five non-overlapping regions (i.e., USA, UK, Japan, Developed Europe ex. UK and, Asia ex. Japan). Considering as investment period the interval from May 2004 to June 2015, our results show that each single factor yields positive premia in exchange for risk, which can lead to considerable underperformance and extensive recovery periods during times of crisis. The best factor investments can be found in Asia ex. Japan and the US. However, risk factor based portfolio construction across the various regions enables the investor to exploit low correlation structures, reducing the overall volatility, as well as tail- and extreme risk measures. Finally, the empirical results point towards the long-only global minimum variance portfolio, as the best risk minimization strategy.

Keywords Risk factors · Minimum risk portfolio · Regularization · Portfolio optimization · Transaction cost

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1 Introduction

Until today, most investors still allocate their funds by directly investing in a universe of various asset classes, like equities or bonds. However, a seemingly well-diversified allocation across many securities can eventually generate a very concentrated portfolio in terms of risk factor exposure. Relying on standard asset pricing models, such as the intertemporal Capital Asset Pricing Model (CAPM) (Merton 1973) or the Arbitrage Pricing Theory (Ross 1976), asset class returns can be explained by their exposure to systematic risk factors. An allocation of funds directly to the individual risk factors, as opposed to securities, can have advantages (Ilmanen and Kizer 2012). First of all, it provides a better risk management mechanism, as investors control ex-ante the factor exposure of their portfolios, as opposed to being passively exposed to the embedded factor tilts. In other words, by building multi-factor portfolios, investors turn risk allocation into "a choice rather than a fate" (Amenc et al. 2015). Secondly, it may represent a passive, cheaper, scalable, and transparent alternative to active asset management. As suggested by Hasanhodzic and Lo (2006), Giamouridis and Paterlini (2010) and Weber and Peres (2013) even factor-based hedge fund replication is, on average, possible. Finally, a direct exposure to the various risk factors, allows to exploit typically lower correlation levels, than those of traditional asset classes.

Although factor investing should be preferred when choosing the portfolios allocation, the investor still needs to decide on how to allocate her wealth in the most effective way. According to the CAPM theory, market-capitalization weighting is efficient for optimal asset allocation. That is why Capitalization-Weighted indexes (CW) have played a central role in the investment industry: both as a starting point for active management and as target point for passive management. However, the CAPM assumptions do not hold in financial markets and the resulting allocations are inefficient (e.g. Haugen and Baker 1991; Hsu 2006). Also standard asset allocation models, such as the mean-variance approach of Markowitz (1952), typically rely on the unrealistic normality hypothesis and are highly sensitive to estimation errors, especially in the expected returns (Michaud 1989), resulting in unattainable asset allocations, leaving ex-ante efficient portfolios rather inefficient ex-post (see i.e. Broadie 1993; DeMiguel et al. 2009b). In this context, the portfolio literature has witnessed a surge of contributions surrounding the recent development of risk budgeting and strategies with special emphasis on risk parity portfolios (Maillard et al. 2010; Roncalli 2013). Such approaches have been developed to overcome the common criticisms of the CW indexes, such as the fact that they overlook correlations across stocks, concentrate only in a few of them or overweight overpriced stocks. In simple words, they rely on the premise that estimation risk is particularly acute for expected returns and less so for correlations and volatilities (Merton 1980; Chopra and Ziemba 1993) and that one way is to simply avoid estimating expected returns altogether and to resort to risk-minimization techniques. Facing this situation, one must first identify the relevant investment opportunities for the investor, and then, determine which optimization scheme to use, to guarantee an efficient risk-return allocation.

The contribution of this paper can be summarized as follows: First of all, we provide an extensive analysis of 5 risk factors (i.e. size, value, momentum, high profitability and low investment) for 5 non-overlapping regions (i.e., the US, the UK, Japan, Developed Europe excluding the UK, and Asia excluding Japan), to point out potential diversification opportunities in multi-factor allocations. We begin our analysis with a conditional performance analysis, discussing the risk and return characteristics of each individual factor during both favorable and unfavorable market states. Second, considering the 25 factors as investment universe, we compare eight popular risk-minimization strategies, including Equal Weighting portfolios (EW), Global Minimum Variance-Long Only portfolios (GMV-LO), Global Minimum Variance under Norm constraints portfolios (GMV-N), Inverse Volatility portfolios (IV), Equal Risk Contribution portfolios (ERC), Minimum Linear Torsion-Risk Parity-Long Only portfolios (MLT-RP-LO), Maximum Decorrelation under Norm constraints portfolios (MDC-N) and Maximum Diversification under Norm constraints portfolios (MDV-N). Our investigation focuses on risk, which is why all of our strategies, except the EW, require as only input an estimate of the covariance matrix. Still, we include the EW, as it is the most diversified portfolio and is considered a tough benchmark to beat. In what follows, we refer to all the approaches, with slight abuse of terminology, as risk-minimization strategies. Our analysis allows us to compare the resulting portfolio allocations with the individual factor performances and discuss the pros and cons of each strategy, with regard to risk and profitability.

Finally, we evaluate risk diversification by means of the diversification ratio and the effective number of correlated and uncorrelated bets (Roncalli 2013; Meucci et al. 2014). To assess the robustness of our results, we perform a rolling window analysis with two window sizes, including the last 2 and 3 years of daily return data. Furthermore, we perform the robust hypothesis test for Sharpe Ratio (SPR) equivalence, following Ledoit and Wolf (2008). Lastly, we investigate the portfolios behavior under different transaction cost regimes, considering no, low and high transaction costs.

For the chosen investment period of May 2004 to July 2015, our results show that all factors yield positive premia, in exchange for risk, which can lead to considerable underperformance and longer recovery periods during crisis times. Furthermore, our conditional performance analysis shows that none of the five factors consistently performs poorly during bad market states. Evaluating individual regions, reveals that the US and Asia excluding Japan offer the most attractive investment opportunities. Still, the investment on a portfolio basis in factors across regions, enables the investor to exploit low correlation structures and reduce her risk exposure. Furthermore, our results show that no portfolio is statistically different from any other portfolio with regard to the SPR, and no approach consistently dominates the other approaches on all measures. For example, our results point towards the GMV-LO as the best strategy for an investor seeking minimum risk, short drawdown periods, and a high certainty equivalent. Although this portfolio is prone to high turnover, its performance persists in no and low transaction cost regimes, while its return is substantially reduced under high transaction costs. Consequently, an investor following risk factor based portfolio construction, is able to select the best strategy based on the preferred individual metrics.

The paper is structured as followed. Section 2 describes the various risk minimization allocation strategies. Section 3 provides an overview of the factors, including descriptive statistics. Section 4 discusses the empirical results from the conditional performance analysis and the risk minimization allocation strategies. Section 5 concludes.

2 Risk-minimization strategies

For our comparison of multi-factor portfolio performance, we consider eight portfolio optimization methods. Such approaches can be considered state-of-the-art in the portfolio management literature and allow us to better point out the advantages and disadvantages of multi-factor investments within a quantitative portfolio management framework. Table 1 summarizes each methodology, listing the main idea, the optimization objective, as well as the main reference. A brief description of each method, including the main advantages and disadvantages, is provided in Appendix 1. For more rigorous derivations, we refer the reader to the references listed in Table 1.

One characteristic common to all of the strategies analysed in this study is that the only input required to determine the portfolio allocation is the assets risk, here quantified by the covariance matrix. Clearly, if the estimated covariance matrix differs from the true value, using it as an input might lead to non-optimal weights. Estimation error can then badly affect the portfolios performance, especially when the ratio $\frac{N}{T}$, in which N is the number of assets and T is the number of observations, is large (Ledoit and Wolf 2004). Coqueret and Milhau (2014) conducted an extensive comparison of 12 commonly used covariance matrix estimators, and evaluated the best estimator, on the basis of the average volatility and turnover of the GMV portfolio, which is notoriously sensitive to the robustness of the covariance matrix. They conclude that the implicit factor model estimators, in which the chosen factors are based on principal component selection with logarithmic criterion (Bai and Ng 2002) or Random Matrix Theory criterion (Daly et al. 2008) dominates the other estimators.

In this paper, we follow Coqueret and Milhau (2014) and estimate the sample covariance matrix using Principal Component Analysis (PCA). We select the number of principal component factors (PCF) considered in the estimation of the correlation matrix, by using the Random Matrix Theory criterion. PCA is a standard procedure for extracting uncorrelated factors from a basket of correlated constituents (Meucci 2010; Deguest et al. 2013; Lohreet al. 2014). By using Random Matrix Theory to select only systematic factors that explain most of the variance in the dataset, the amplitude of the off-diagonal elements of the covariance matrix is reduced. Due to space limitations, we refer the reader to Coqueret and Milhau (2014) and references therein for a detailed explanation of covariance estimation using Principal Component Analysis and Random Matrix Theory.

3 Data

When selecting a risk factor, the investor should avoid the pitfalls of non-persistent factor premia and should try to achieve a robust performance. For that she needs to: (1) have a sound economic rationale for the existence and persistence of a positive premium, which reduces the likelihood of these factors being significant in-sample by pure chance; and (2) be wary of complex and proprietary factor definitions, resulting from data-mining, and hence should stick to simple factor definitions that are widely used in the literature (Gelderen and Huij 2013).

This study focuses on five of the most popular risk factors that satisfy these conditions and have therefore attracted widespread interest from both academics and practitioners: size, value, momentum, profitability, and investment. Please refer to “Appendix 2” for a detailed description of the individual factors. We consider these indices for five non-overlapping regions, respectively, including the US, the UK, Japan, Developed Europe ex-UK, and Asia Pacific ex-Japan, leading to a universe of 25 factor indices. All benchmarks are total return indices, with dividends reinvested, and the currency of denomination is US dollars. Daily return data was taken from EDHEC Risk Institutes (ERI), covering the period from 21 June 2002 to 30 June 2015 (3397 daily observations). As we want to compare the individual risk factors with our portfolios, we align the reported performance of the risk factors with the first date, on which we initiate our portfolio. We refer to this period as the investment horizon, which covers the period from 21 May 2004 to 05 June 2015 (2898 observations).

Table 1 Overview of risk-minimization strategies

Strategy	Idea	Optimization	References
EW	Naïve weight diversification	$w_i = \frac{1}{N}, \forall\, i = 1, \ldots, N$	Benartzi and Thaler (2001), Windcliff and Boyle (2004), DeMiguel et al. (2009b)
GMV-LO	Equal marginal risk contribution	$\min\limits_{\boldsymbol{w} \in \mathbb{R}^N} \sigma_p^2 = \boldsymbol{w}' \Sigma \boldsymbol{w}$	Markowitz (1952), Jagannathan and Ma (2003)
GMV-N	ℓ_2-Regularized equal marginal risk contribution	$\min\limits_{\boldsymbol{w} \in \mathbb{R}^N} \sigma_p^2 = \boldsymbol{w}' \Sigma \boldsymbol{w}$ $s.t.\ \ \|\|\boldsymbol{w}\|\|_2^2 \leq \frac{3}{N}$	Markowitz (1952), Clarke et al. (2011)
IV	Inverse volatility based weights	$w_i = \frac{\frac{1}{\sigma_i}}{\sum_{i=1}^N \frac{1}{\sigma_i^2}}, \forall\, i = 1, \ldots, N$	Maillard et al. (2010)
ERC	Equal risk contribution	$\min\limits_{\boldsymbol{w} \in \mathbb{R}^N} \sum_{i=1}^N \left(\frac{w_i\ (\Sigma \boldsymbol{w})_i}{\sigma_p^2} - \frac{1}{N} \right)^2$	Maillard et al. (2010), Bruder and Roncalli (2012), Roncalli (2013)
MLT-RP-LO	Equal risk factor contribution	$\min\limits_{\boldsymbol{w} \in \mathbb{R}^N} \sum_{i=1}^N \left(\frac{v\{w_{F_k} r_{F_k}\}}{v\{r_p\}} - \frac{1}{N} \right)^2$	Meucci et al. (2014)
MDV-N	ℓ_2-Regularized equal correlation contribution	$\max\limits_{\boldsymbol{w} \in \mathbb{R}^N} DR(\boldsymbol{w}) = \frac{\sum_{i=1}^N w_i \sigma_i}{\sqrt{\boldsymbol{w}' \Sigma \boldsymbol{w}}}$ $s.t.\ \ \|\|\boldsymbol{w}\|\|_2^2 \leq \frac{3}{N}$	Choueifaty et al. (2011)
MDC-N	ℓ_2-Regularized correlation structure	$\min\limits_{\boldsymbol{w} \in \mathbb{R}^N} C(\boldsymbol{w}) = \boldsymbol{w}' \Omega \boldsymbol{w},$ $s.t.\ \ \|\|\boldsymbol{w}\|\|_2^2 \leq \frac{3}{N}$	Christoffersen et al. (2010)

The table summarizes the eight risk-minimization strategies. We report from left to right for each optimization, the strategies acronym (Strategy), the underlying Idea (Idea), the optimization objective (Optimization), and the references (References), to which we redirect the reader for an in depth analysis. The risk-minimization strategies comprise the equal weighting portfolio (EW), the global minimum variance long only portfolio (GMV-LO), the global minimum variance under norm constraints portfolio (GMV-N), the inverse volatility portfolio (IV), the equal risk contribution portfolio (ERC), the minimum linear torsion-risk parity-long only portfolio (MLT-RP-LO), the maximum decorrelation under norm constraints portfolio (MDC-N) and the maximum diversification under norm constraints portfolio (MDV-N). All optimization problems require either the covariance matrix (Σ) or the correlation matrix (Ω) of the return factor time series as an input, and need to satisfy the no-short selling constraint ($w_i \geq 0$) and the budget constraint ($\sum_i^N w_i = 1$), where N is the number of risk factors and $\boldsymbol{w} = [w_1, w_2, \ldots, w_N]'$ is the portfolio weight vector. Please refer to "Appendix 3", for detailed description of each strategy and corresponding notation

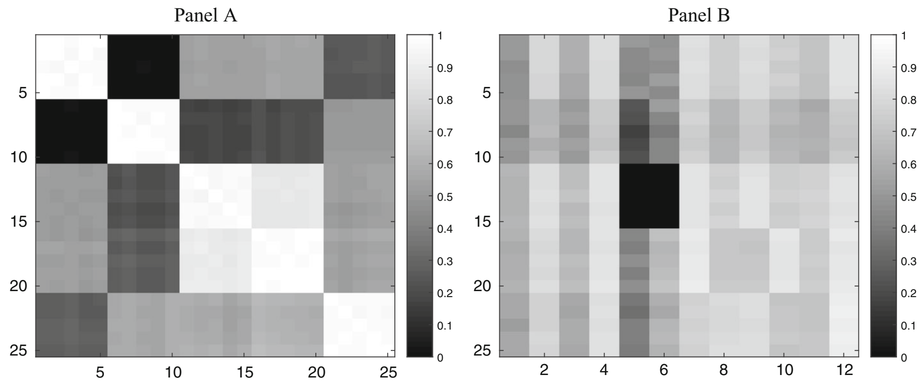

Fig. 1 Factor indices correlations across regions and regimes. The figure shows in *Panel A* the correlation coefficients among each of the 5 factor indices sorted by the 5 non-overlapping regions considered. The factors are in the order of appearance *size*, *value*, *momentum*, *investment* and *profitability*, reported for the USA, Japan, the UK, Developed Europe ex-UK, and Asia Pacific ex-Japan, respectively, leading to a total of 25 factors. The values above the diagonal are calculated based on the entire investment horizon (May 2004–June 2015), while the figures below the diagonal are calculated based on the global financial crisis period (December 2007–June 2009, based on the US Business Cycle Expansions and Contractions dates as published by NBER Business Cycles). *Panel B* reports the average correlation coefficient for each of the five factor indices, conditional on the respective market regime and state, further differentiated by the five non-overlapping regions. The x-axis represents the different market regimes and states, as defined in "Appendix 4". Reported are, from left to right, *Bull Market Regime*, *Bear Market Regime*, *Low Volatility Regime*, *High Volatility Regime*, *Low Credit Risk Regime*, *High Credit Risk Regime*, *High Interest Rate Regime*, *Low Interest Rate Regime*, *Positive Economic Surprises*, *Negative Economic Surprises*, *Favorable Economic Conditions* and *Unfavorable Economic Conditions*, respectively. The y-axis represents the five non-overlapping regions, starting from the top, represented by the USA, Japan, the UK, Developed Europe ex-UK, and Asia Pacific ex-Japan, respectively

ERI Scientific Beta constructs the different factor indices by selecting and segregating stocks based on straightforward scoring criteria, e.g. the free float market cap for the mid size factor or the total asset growth over past two years for the low investment factor, to avoid the risk of complex and unproven factor definitions, resulting from data-mining.[1]

Panel A of Fig. 1 shows the correlation coefficients between the 25 factor-tilted indices, split into the 5 non-overlapping regions, respectively. The values above the diagonal are calculated based on the entire investment horizon (21 May 2004–5 June 2015) while those below the diagonal are based on the global financial crisis period (03 December 2007–01 June 2009). As it is evident from the light color in Panel A, the correlations between the factors in any given region (main diagonal elements) are very close to unity, while the values are much lower when looking at the cross-country correlations (off main diagonal elements). Therefore, we expect some diversification benefits to be gained from investing across factors within a given region, as the investor may face correlation coefficients that are high but still below one. This holds for the full sample, and the crisis period.

Furthermore, the investor can also expect considerable diversification benefits from investing across factors and across the different regions, as shown in Panel A. Not surprisingly, the latter mostly represents values that are lower than the within-country correlations. Finally, when looking at the different time periods separately, we can see that there is no pronounced difference in correlation values between the crisis period and the full sample period. As

[1] Please refer to the ERI Scientific Beta Website for a complete list of all scoring criteria. http://www.scientificbeta.com/.

a result, factor diversification benefits do exist independently from the market regime and investors can count on the low correlation between factor indices to persist, especially when they need it most, i.e. during periods of market distress. The results above are in line with the findings of Ilmanen and Kizer (2012), who show that factor diversification for long-only investors is more effective than the traditional asset-class diversification methods.

4 Empirical results

4.1 Conditional performance analysis

We complement our analysis of the factor-tilted indices, by introducing a conditional performance analysis, and considering the performance of the risk factors in two states of the world, namely *Favourable* and *Unfavourable Market Conditions*. This enables us to identify the preferred investment universe for each of the two economic environments. Asset pricing theory suggests that positively rewarded factors perform poorly during bad times, and more than compensate during good times so that they generate an average positive excess return across all possible market conditions. Conditional performance analysis helps the investors to make better informed decisions with respect to the factor tilts to which their portfolio should be exposed to. For example, an unconditional evaluation of the factor index performance made during bearish markets will not yield a useful estimate of the forward performance, if the next period is bullish (Ferson and Qian 2004).

In what follows, we first analyze whether the average correlation of the factors is higher in any of the two states of the economy and then test, if we can reject the hypothesis that the average return, average volatility, and the corresponding reward-to-risk-ratio are equal in both states of the world at the 99% confidence level. In order for the factors to be risk factors, they should perform poorly in bad states of the world and more than compensate in good states of the world.

As a formal definition of "bad state" versus "good state" of the world does not exist, we propose to use six variables, namely market returns, realized one month volatility, interest rate term spread, Credit Default Swap (CDS) spread, Economic Surprises Index, and Economic Conditions Index to define Unfavorable versus Favorable Market Conditions. "Appendix 4" presents the economic justification for the definition of Favorable and Unfavorable states given the above economic indicators.

Panel B of Fig. 1 shows the average correlation between a given index and the remaining 24 indices, conditional on one of the 12 Favorable or Unfavorable Market Conditions, respectively. The average correlation does not always increase during Unfavorable Market Conditions. For example, the average correlation of almost all factor-tilted indices decreases during Low Interest Rate and Negative Economic Surprise regimes, thus providing increased diversification opportunities. Also, within most of the five regions, the average correlation increases during Unfavorable Market Conditions, as proxied by the Bear market regime, High Volatility regime, High Credit Risk regime, and Unfavorable Economic Conditions regime. Nevertheless, the average correlation is below 0.85. As a result, portfolios combining all factor-tilted indices can still exploit the risk reduction effect stemming from less than perfect correlation, even during Unfavorable Market Conditions. The low correlations under both Favorable and Unfavorable Market Conditions confirm that the cyclicity of returns differs from one factor and region to the other. Hartmann et al. (2001) show that during market crashes, the diversification benefits provided by traditional asset classes and investments in different geographic locations are disappointing, as assets returns become very correlated. This leads to co-crashes between stock and bond markets and between developed and emerging economies, which supports our findings.

Table 5 in "Appendix 4" reports the return, volatility, and reward-to-risk of the five indices for each region and for the twelve different market states. The first Panel in Table 5 focuses on US factors performance. Across all factors, average, the monthly return is negative (−1.72%) during Bear Market regimes and positive (3.91%) during Bull Market regimes, with the difference in means between the two being significantly different from zero. Furthermore, their average volatility is much higher during the High Volatility regime (6.41%) than during the Low Volatility regime (2.85%). This implies that the US Factors are not immune to increasing market volatility conditions. However, it is interesting to note that in all cases, except for the momentum index, the factor-tilted indices deliver positive return regardless of the volatility regime. Moving on to the Credit Risk regime analysis, all factors suffer from negative performance during High Credit Risk regime periods (average return of −0.23%), while delivering high positive returns during Low Credit Risk regime periods (average return of 2.98%). Their volatility stays relatively constant in both market states (average volatility of 3.62%). It is worth noting that all factors carry similar and positive returns during the two Interest Rate, and Economic Surprise regimes. Finally, although the volatility of all factors increases significantly during Unfavorable Economic Conditions (average volatility of 6.28%) compared to Favorable Economic Conditions (average volatility of 3.81%), their returns remain positive in both states (except for the momentum index).

The last 4 Panels of Table 5 display the risk adjusted performance of the factor-tilted indices during Favorable and Unfavorable Market Conditions in the remaining four regions (Japan, UK, Developed Europe ex-UK, and Asia ex-Japan). In what follows, only patterns different from those observed for the US factor-tilted indices are highlighted.

Firstly, for the Europe ex-UK factor indices, the two states of the Interest Rate regime and Economic Surprises regime lead to differences in volatility and return, respectively. The Size, Value, and Profitability indices are characterized by significantly lower volatility during Low Interest Rate regimes, typically indicating recessions, while delivering similar and positive returns in both states (Low and High Interest Rate regimes). Unlike the factors in all other regions, the Developed Europe ex-UK factors incur negative returns during periods of Negative Economic Conditions (average return of −0.32%) and positive returns during periods of Positive Economic Conditions (average return of +2.28%). Secondly, during the High Volatility regime, factor indices in Japan, Developed Europe ex-UK and Asia ex-Japan exhibit both: significantly higher volatility (6.47, 7.38, and 6.68%, respectively) and significantly lower return (−0.28, −0.52, and −0.23%, respectively) compared to the volatility (3.96, 3.65, and 3.02%, respectively) and return (+1.30, +2.57, and +2.62%, respectively), experienced during the Low Volatility regime. For the US and UK factor indices, although they mirror the movement in market volatility, their returns remain positive. Thirdly, in all five regions, Unfavorable Economic Conditions translate into higher factor index volatility. In addition, for the factor indices based in Japan, UK, Developed Europe ex-UK, and Asia ex-Japan, returns are less resilient to the change in market volatility, and turn negative.

Summing up, these findings reveal that none of the factors in different regions consistently performs unfavorably in bad states of the world. More importantly, when they do underperform in bad states of the world, all factor indices more than compensate in Favorable Market States. This conclusion complements the findings of Amenc et al. (2015). In their conditional performance analysis, focusing only on US factor-tilted indices, the authors use, among others, market returns, unemployment rate, inflation rate, and the Industrial Production Index to differentiate between the different states of the world.

4.2 Return and risk analysis

We compare the performance and risk of the 25 factors and the eight portfolio strategies estimating the input parameters with a rolling window of 500 (2 years) and 750 (3 years) daily observations, and rebalance the optimal portfolios every 30 days over the period June 2002–June 2015. With a window size of 500, our portfolio is initiated the first time in May 2004 and is rebalanced every 30 days until June 2015.[2] This provides us with 96 and 88 out-of-sample observations, respectively. To align the performance of our individual risk factors with those of our portfolios, we compute their return and risk properties during the period May 2004–June 2015, excluding the first 500 (750) observations used for finding the optimal portfolio weights.

Table 2 displays the out-of-sample return and risk measures for the various factor-tilted indices and regions, and for the eight portfolio optimization strategies. Reported are the annual returns, the annual volatilities, the annual loss standard deviations, and the SRP and Sortino Ratio (STR). A detailed description of the various measures used can be found in "Appendix 3". If not stated otherwise, the drawn conclusions are robust to the change in the window size and we report the results for the estimation period comprising the last 3 years of data in parentheses.

Analyzing the values for the individual factor-tilted indices, given in the first five Panels of Table 2, it is easy to see that all chosen factors positively reward the investor over the considered time period. Furthermore, the risk-adjusted performance across the factor-tilted indices within each region is mostly similar. The most heterogeneous region is the UK, which shows the highest differences among the SRP and STR. Here, the preferred investment is the Profitability factor, while Value investment underperforms. Across regions, however, the performance differs substantially. For example, Asia ex. Japan and US regions stand out, with an above average SPR of 0.64 (0.49) and 0.62 (0.59), as well as a STR's of 0.79 (0.62) and 0.74 (0.71) for a window size of 500 (750) observations. The laggard region, for $WS = 500$, is Japan, reporting an average SPR of 0.41 and STR of 0.56, while for a $WS = 750$, Europe excluding UK underperforms with a SPR of 0.36 and a STR of 0.44 (each value computed as an average across the five factor-tilted indices). Extending the sample size to a time period of 3 years, the overall returns decrease while the volatility increases. This is not surprising, as with the longer time interval, we now have more out-of-sample observations, which include data from the financial crisis of 2007–2009.

Looking at the risk-minimization strategies in the last Panel, we notice that all multi-factor portfolios have returns that are in line with the average return of the constituents, which is 9.8% (8.5%). However, there is a substantial reduction in risk, as the average volatility of the constituents is around 18.7% (19.2%), whereas the volatility of the multi-factor portfolios is between 13.9% (14.9%) and 16.6% (17.6%). The average SPR and STR across the 25 factor-indices, represented by the EW portfolio, are 0.59 (0.49) and 0.71 (0.59), respectively, whereas the average SPR and STR across the multi-factor portfolios are 0.63 (0.51) and 0.75 (0.61). Furthermore, our results show that the GMV-LO is the best performing portfolio with regard to the SPR (0.68 (0.54)) and the SRT (0.82 (0.64)). It reports a variance of 13.93% (15.15%), while the EW reports the highest volatility at 16.58% (17.56%). Among

[2] We extend our analysis to also cover, 250 (1 year) and 1250 (5 years) daily observations. The reported results with regard to the factor indices and the portfolios are robust towards those changes, and available upon request.

Table 2 Return and risk analytics

	WS = 500					WS = 750				
	Ret. (%)	Vol. (%)	LSD (%)	SPR	STR	Ret. (%)	Vol. (%)	LSD (%)	SPR	STR
US										
Size	10.57	17.02	18.67	0.62	0.73	10.05	17.26	18.52	0.58	0.69
Value	10.36	16.22	17.89	0.64	0.76	9.34	16.58	17.09	0.56	0.68
Momentum	9.32	16.47	18.23	0.57	0.66	8.59	16.74	17.71	0.51	0.61
Investment	10.75	15.79	17.55	0.68	0.80	10.28	15.81	16.59	0.65	0.78
Profitability	10.30	16.08	17.67	0.64	0.75	10.33	15.77	16.24	0.66	0.78
Japan										
Size	5.66	15.04	9.15	0.38	0.52	4.76	14.59	11.07	0.33	0.43
Value	6.63	15.37	9.01	0.43	0.59	5.35	14.60	10.76	0.37	0.48
Momentum	5.26	15.43	9.84	0.34	0.46	4.45	14.98	11.08	0.30	0.39
Investment	6.41	14.92	9.59	0.43	0.58	5.44	14.45	10.28	0.38	0.50
Profitability	6.91	14.20	9.27	0.49	0.66	6.59	15.05	12.20	0.44	0.57
UK										
Size	8.68	19.69	19.37	0.44	0.53	7.87	19.46	16.85	0.40	0.50
Value	7.62	20.88	20.45	0.37	0.44	6.49	21.09	18.95	0.31	0.38
Momentum	10.37	19.03	18.66	0.54	0.65	9.52	18.23	16.99	0.52	0.64
Investment	9.43	18.63	17.64	0.51	0.61	8.75	18.48	15.71	0.47	0.59
Profitability	10.50	17.83	17.65	0.59	0.70	9.95	17.91	16.17	0.56	0.68
Europe ex-UK										
Size	8.74	21.22	18.81	0.41	0.50	7.06	22.53	19.67	0.31	0.39
Value	8.09	23.94	20.91	0.34	0.42	6.08	25.88	22.29	0.24	0.29
Momentum	10.23	20.49	18.65	0.50	0.60	8.54	21.50	19.62	0.40	0.49
Investment	10.03	20.75	17.66	0.48	0.59	8.60	22.09	18.73	0.39	0.49
Profitability	10.51	19.36	17.09	0.54	0.66	9.24	20.78	18.19	0.44	0.55

Table 2 continued

	WS = 500					WS = 750				
	Ret. (%)	Vol. (%)	LSD (%)	SPR	STR	Ret. (%)	Vol. (%)	LSD (%)	SPR	STR
Asia ex-Japan										
Size	14.01	22.65	20.37	0.62	0.76	11.35	24.57	20.93	0.46	0.59
Value	14.11	23.19	20.53	0.61	0.75	11.06	24.99	20.56	0.44	0.57
Momentum	15.11	22.93	20.72	0.66	0.81	12.41	24.55	20.78	0.51	0.65
Investment	14.06	19.98	17.90	0.70	0.86	11.53	21.72	18.01	0.53	0.67
Profitability	12.35	19.96	18.26	0.62	0.76	10.68	22.02	18.20	0.49	0.62
Risk minimization strategies										
EW	9.84	16.58	15.60	0.59	0.71	8.57	17.56	16.48	0.49	0.59
GMV-LO	9.45	13.93	12.79	0.68	0.82	8.14	15.15	15.19	0.54	0.64
GMV-N	9.20	16.42	15.57	0.56	0.67	8.55	17.26	16.28	0.50	0.60
IV	9.87	16.34	15.36	0.60	0.72	8.56	17.41	16.48	0.49	0.59
ERC	9.75	15.75	14.49	0.62	0.74	8.42	16.90	16.14	0.50	0.60
MLT-RP-LO	10.02	15.41	14.17	0.65	0.78	8.52	16.63	15.90	0.51	0.62
MDC-N	9.29	14.17	12.85	0.66	0.79	8.04	15.08	14.62	0.53	0.64
MDV-N	9.22	13.98	12.52	0.66	0.79	7.97	14.94	14.69	0.53	0.64

The table reports core return and risk statistics for the factor-tilted indices stated by region, as well as for the eight risk-minimization strategies. All values representing the investment period from May 2004 to June 2015, considering a rolling window approach of WS = 500 and WS = 750, respectively. Stated from left to right are the annual return (Return), the annual volatility (Vol.), the annual loss standard deviation (LSD), considering 252 days per year, as well as the Sharpe (SRP) and Sortino (STR) Ratio. The risk-free rate for the Sharpe and Sortino ratio is given by the 3-month US Treasury Bills yield. The performance ratios that involve average returns are based on the geometric average, such as to reflect the multiple holding period returns for investors. The considered factor-tilted indices include size, value, high momentum, low investment and high profitability for the following five regions, respectively: USA, Japan, UK, Developed Europe ex-UK, and Asia Pacific ex-Japan. The risk-minimization strategies comprise the equal weighting portfolio (EW), the global minimum variance long only portfolio (GMV-LO), the global minimum variance under norm constraints portfolio (GMV-N), the inverse volatility portfolio (IV), the equal risk contribution portfolio (ERC), the minimum linear torsion-risk parity-long only portfolio (MLT-RP-LO), the maximum decorrelation under norm constraints portfolio (MDC-N) and the maximum diversification under norm constraints portfolio (MDV-N) the risk parity portfolios, the IV and ERC strategies have risk profiles that are inferior when compared to the more sophisticated methods like the MDC-N, MDV-N and MLT-RP-LO. Here, the MDC-N and MDV-N perform best.

The performance of the EW is not surprising, as the naïve strategy merely disregards any correlation structure and can thus not profit from any diversification benefits. The observation that the GMV-LO stands out can be linked to the large literature on estimation errors and robust portfolio optimization (see i.e. Best and Grauer 1991; Brodie et al. 2009; Becker et al. 2015; Platanakis and Sutcliffe 2017). In fact, the original mean-variance optimization framework developed by Markowitz (1952), is highly sensitive towards the type of input parameters and the presence of estimation error. Michaud (1989) refers to this problem as the "error maximization", i.e. the fact that the mean and the covariance matrix, as the two input parameters, are prone to estimation errors, which have a large effect on the estimation of portfolio weights. To circumvent these problems, the financial literature started to rely on more sophisticated statistical methods. Among the most successful tools, are robust input estimation and regularization methods. Robust methods work by trying to directly improve the input parameters using for example shrinkage techniques (see e.g. Ledoit and Wolf 2004; Fan et al. 2008, 2009). Regularization methods, instead, typically work by adding additional weight constraints, such as the constraint on the ℓ_1-Norm[3] of the portfolio weights to the optimization problem (see e.g. DeMiguel et al. 2009b; Fan et al. 2012; Fastrich et al. 2015).

With regard to the GMV-LO, Jagannathan and Ma (2003) show that adding weight constraints to the global minimum variance portfolio, reduces estimation errors associated with the covariance matrix estimator, and improves the portfolio performance. More recently, Fan et al. (2012) provide some theoretical support for the findings of Jagannathan and Ma (2003). They show that in the GMV framework, adding a no short selling constraint combined with the budget constraint, and restricting the added norm to be equal to 1, is a special case of ℓ_1-regularization. Consequently, the theoretical results provided in Fan et al. (2012), and specifically the fact that ℓ_1-regularization reduces the maximum estimation error, associated with the largest component in the covariance matrix, which subsequently results in better out-of-sample properties, apply in this case. This allows the investor to better control for the impact of estimation error in the input parameters.

The other strategies, like the GMV-N, MDV-N or MDC-N, also consider a regularization term, by adding the ℓ_2-norm to the optimization, which is known as ridge regression analysis in the statistical literature. However, the ℓ_2-norm does not promote sparsity, by setting some asset weights equal to zero, but rather allows to better deal with multicollinearity. Finally, comparing the optimization problems in Table 1, we also notice that the MLT-RP-LO relies on target risk minimization by using a transformed function of the covariance input estimates. Thus, it is paying a price in terms of volatility minimization for increased diversification when measured in terms of ENCB and when compared to the GMV-LO. This makes the GMV-LO the best performing strategy.

We complement our analysis with the robust hypothesis test for the SPR developed by Ledoit and Wolf (2008). As such, we test if there is a significant difference between the highest SPR, given by the GMV-LO and the SPR of any other portfolio. As none of our results show that there is a significant difference between the portfolios at the 1, 5 and 10% levels, we do not report the p-values here, but make them available upon request. Do these results indicate that the strategies can be considered equivalent from an investors viewpoint? The statistical tests only suggest that the SPR are comparable. Still, from the table we can see that all portfolios provide different return and risk characteristics. Furthermore, investors

[3] The ℓ_1-Norm is defined as: $||\boldsymbol{w}||_1 = \sum_{i=1}^{K} |w_i|$.

will have different preferences depending on what they want to accomplish. Consequently, an investor, who wants to minimize the risk while disregarding the return, can identify the strategy that will provide her with the lowest possible volatility.

Summing up, our results show that individual risk factor investing might be promising with regard to returns, but can expose the investor to high uncertainty. Methods like the EW, IV, or ERC show lower variance, when compared to individual factors, but underperform with regard to more sophisticated portfolio methods like the MDC-N or MDV-N. The best risk-return trade off is attained when following a GMV-LO strategy.

4.3 Tail-risk and diversification analysis

As measuring risk can have many dimensions, besides volatility and loss standard deviation, we focus below on quantifying tail risk. Table 3 reports the Tail- and Extreme Risk measures for the individual factor-tilted indices, and for the eight risk minimization allocation strategies. In particular, we consider the Maximum Drawdown (MDD), the Maximum Drawdown Period in years (Period), the Recovery Period after the Maximum Drawdown in years (Rec.), as well as the historical Value at Risk (VaR 5%) and Expected Shortfall (ES 5%), both evaluated at the 5% significance level. A detailed description of each measure can be found in "Appendix 3".

The first four Panels of Table 3 state the extreme risk statistics for the individual factor-tilted indices. Considering two years of data, all factor indices, except those for Japan, exhibit more severe MDDs, VaR, and ES. This result also confirms the values of Table 2. Japan stands out as a region with low returns and low volatility. As such, any market downturn might be lower compared to other regions, but it will also take the investor on average longer to overcome this valley. This is supported by nearly twice as high values for the MDD Period and an above average recovery time of roughly 3 years. Excluding Japan, among the various regions, the best performing areas are Asia ex-Japan and the US. Although the former shows comparably equal and sometimes even higher MDDs, the good performance stems from the short period over which a market downturn occurs and the comparably reduced time of only 1 year that the investor needs to wait for a recovery. The US shows one of the lowest MDDs, the shortest Period over which the MDD occurs, and the fastest recovery rates. All observations are robust to changes in the window size.

Considering the portfolio allocation strategies in Panel 5 of Table 3, and excluding Japan, all portfolios have Tail- and Extreme Risks, which are equal to or better than those of the individual factors. The majority of VaR 5% and ES 5% figures are higher at the index level, than at the portfolio level. In some cases, this difference is not negligible: the average VaR 5% and ES 5% are above 12 and 18% for the factor indices in Developed Europe ex-UK and Asia ex-Japan, respectively, compared to the average VaR 5% and ES 5% of the portfolios at 9% and 13%.

All portfolios experience their MDD approximately during the period October 2007–March 2009, which overlaps with the recent global financial crisis. However, their recovery time is very different. The EW portfolio needs around four years to recover the loss suffered during the MDD period. This is almost double the time that any of the GMV and Risk Parity portfolios need, and it is in line with the time needed by the MDC-N and MLT-RP-LO portfolio. The lowest MDD is achieved by the MDV-N strategy (52.71%), closely followed by the GMV-LO (53.63%).

As in Sect. 4.2, we can observe that the naive allocation procedures perform comparably worse when considering the two sample periods. Interestingly the GMV-N is the worst

Table 3 Tail- and extreme-risk analytics

	WS = 500					WS = 750				
	MDD (%)	Period	Rec.	VaR 5%	ES 5%	MDD (%)	Period	Rec.	VaR 5%	ES 5%
US										
Size	53.42	1.75	1.98	−10.39	−17.38	53.42	1.75	1.98	−8.89	−15.90
Value	53.75	1.51	2.06	−10.99	−16.71	53.75	1.51	2.06	−7.50	−16.02
Momentum	53.25	1.33	3.69	−9.55	−16.74	53.25	1.33	3.69	−8.44	−15.90
Investment	50.82	1.75	1.95	−10.58	−16.00	50.82	1.75	1.95	−7.09	−15.31
Profitability	47.58	1.19	2.09	−9.78	−16.03	47.58	1.19	2.09	−6.99	−14.37
Japan										
Size	46.25	3.04	4.45	−8.78	−10.05	46.25	3.04	4.45	−7.42	−12.30
Value	39.91	3.04	4.41	−7.78	−10.20	39.91	3.04	4.41	−8.79	−12.73
Momentum	44.53	3.04	5.67	−9.13	−11.09	44.53	3.05	5.67	−7.21	−12.69
Investment	36.97	3.04	4.44	−8.62	−10.72	36.97	3.05	4.44	−8.30	−11.81
Profitability	42.77	3.04	2.50	−8.30	−9.92	42.77	3.04	2.50	−7.58	−13.23
UK										
Size	64.45	1.92	4.48	−9.28	−17.63	64.45	1.92	4.47	−10.69	−17.31
Value	67.45	1.45	4.94	−10.95	−18.59	67.45	1.45	4.94	−12.43	−19.69
Momentum	60.10	1.45	4.42	−10.87	−16.38	60.10	1.45	4.42	−11.79	−16.06
Investment	61.72	1.45	4.15	−10.19	−16.49	61.72	1.45	4.15	−11.69	−16.37
Profitability	57.98	1.45	3.17	−10.74	−15.81	57.98	1.45	3.17	−10.41	−16.20
Europe ex-UK										
Size	60.12	1.77	4.89	−12.38	−19.40	60.12	1.77	4.89	−13.93	−20.04
Value	65.72	1.45	5.35	−14.84	−21.44	65.72	1.45	5.34	−17.81	−23.83
Momentum	57.73	1.45	4.68	−12.95	−18.51	57.73	1.45	4.68	−14.22	−19.57
Investment	55.65	1.45	2.23	−13.20	−18.19	55.65	1.45	2.23	−14.00	−19.56
Profitability	54.55	1.45	2.21	−11.74	−17.11	54.55	1.45	2.21	−13.28	−18.46

Table 3 continued

	WS = 500					WS = 750				
	MDD (%)	Period	Rec.	VaR 5%	ES 5%	MDD (%)	Period	Rec.	VaR 5%	ES 5%
Asia ex-Japan										
Size	66.86	1.06	4.48	−12.19	−18.45	66.86	1.06	4.48	−12.39	−20.34
Value	64.28	1.13	2.53	−13.07	−19.66	64.28	1.13	2.53	−12.19	−21.32
Momentum	66.29	1.06	2.93	−12.54	−18.81	66.29	1.06	2.92	−11.35	−20.08
Investment	58.21	1.13	2.25	−11.20	−16.99	58.21	1.13	2.26	−10.94	−18.18
Profitability	60.56	1.06	2.17	−11.00	−16.40	60.56	1.06	2.16	−11.57	−18.23
Risk minimization strategies										
EW	56.55	1.45	4.09	−10.11	−15.24	56.55	1.45	4.09	−10.39	−16.23
GMV-LO	53.63	1.35	2.66	−8.88	−12.40	54.72	1.38	4.65	−8.89	−14.43
GMV-N	57.43	1.45	2.29	−10.12	−15.27	56.19	1.45	4.09	−10.30	−15.80
IV	55.43	1.45	2.27	−9.95	−15.03	55.83	1.45	2.29	−10.42	−16.14
ERC	54.72	1.45	2.29	−9.72	−14.33	55.02	1.45	4.04	−10.03	−15.76
MLT-RP-LO	54.45	1.45	4.29	−9.78	−14.03	54.26	1.45	4.25	−9.80	−15.55
MDC-N	54.93	1.45	4.09	−9.61	−12.45	54.59	1.45	4.25	−8.84	−14.30
MDV-N	52.71	1.74	3.36	−9.34	−12.28	53.10	1.77	4.11	−8.73	−14.19

The table reports extreme risk statistics for the factor-tilted indices stated by region, as well as for the eight risk-minimization asset allocation strategies. All values representing the investment period from May 2004 to June 2015, considering a rolling window approach of WS = 500 and WS = 750, respectively. Stated are the Maximum Drawdown (MDD), the Maximum Drawdown Period in years (Period), the Recovery Period after the Maximum Drawdown in years (Rec.), as well as the historical Value at Risk (VaR 5%) and Expected Shortfall (ES 5%), both evaluated at 5%. The considered factor-tilted indices include mid-size, value, high momentum, low investment and high profitability for the following five regions, respectively: USA, Japan, UK, Developed Europe ex-UK, and Asia Pacific ex-Japan. The risk-minimization strategies comprise the equal weighting portfolio (EW), the global minimum variance long only portfolio (GMV-LO), the global minimum variance under norm constraints portfolio (GMV-N), the inverse volatility portfolio (IV), the equal risk contribution portfolio (ERC), the minimum linear torsion-risk parity-long only portfolio (MLT-RP-LO), the maximum decorrelation under norm constraints portfolio (MDC-N) and the maximum diversification under norm constraints portfolio (MDV-N)

Table 4 Risk diversification analytics of risk-minimization strategies

	EW	GMV-LO	GMV-N	IV	ERC	MLT-RP-LO	MDC-N	MDV-N
WS = 500								
DR	1.37	1.45	1.39	1.38	1.41	1.42	1.48	1.49
ENC	25.00	5.08	24.11	24.46	23.56	10.29	8.15	8.22
ENCB	23.96	9.50	24.00	24.56	25.00	12.02	10.48	10.92
ENUB	23.82	20.65	24.05	24.19	24.68	24.90	21.67	21.69
CEQ	0.03	0.05	0.02	0.03	0.04	0.04	0.04	0.04
Active Pos.	25.00	21.32	24.97	25.00	25.00	14.90	15.70	16.39
Turnover	0.00	0.26	0.08	0.02	0.02	0.13	0.21	0.20
WS = 750								
DR	1.36	1.45	1.38	1.37	1.41	1.41	1.47	1.48
ENC	25.00	4.94	24.07	24.57	23.63	10.16	8.18	8.22
ENCB	24.00	9.05	24.11	24.57	25.00	11.93	10.75	11.08
ENUB	23.85	20.61	24.14	24.22	24.70	24.92	21.82	21.58
CEQ	0.01	0.02	0.01	0.01	0.01	0.02	0.02	0.02
Active Pos.	25.00	21.13	24.97	25.00	25.00	14.66	15.93	16.67
Turnover	0.00	0.20	0.09	0.01	0.01	0.10	0.17	0.16

The table reports risk diversification analytics for the eight risk-minimization asset allocation strategies. All values representing the investment period from May 2004 to June 2015, considering a rolling window approach of WS = 500 and WS = 750, respectively. Stated are the Diversification Ration (DR), the Effective Number of Constituents (ENC), the Effective Number of Correlated Bets (ENCB), the Effective Number of Uncorrelated Bets (ENUB), the certainty equivalent using a risk aversion parameter of $\gamma = 5$ (CEQ), the number of active positions (Active Pos.), and the turnover (Turnover), the latter being defined as the absolute change in the weights of the portfolio, from one period to the other. The risk-minimization strategies comprise the equal weighting portfolio (EW), the global minimum variance long only portfolio (GMV-LO), the global minimum variance under norm constraints portfolio (GMV-N), the inverse volatility portfolio (IV), the equal risk contribution portfolio (ERC), the minimum linear torsion-risk parity-long only portfolio (MLT-RP-LO), the maximum decorrelation under norm constraints portfolio (MDC-N) and the maximum diversification under norm constraints portfolio (MDV-N)

performing portfolio in terms of the MDD, VaR 5%, and ES 5%, considering the last 2 years of data.

Table 4 reports risk diversification analytics, considering the Diversification Ratio (DR), the Effective Number of Constituents (ENC), the Effective Number of Uncorrelated Bets (ENUB), the Effective Number of Correlated Bets (ENCB), the Certainty Equivalent (CEQ), the Turnover (Turnover), and the average number of active positions, across all out of sample observations (Active Pos.).

The information completes the results from Tables 2 and 3. First of all, it is worth noting that for both window sizes, all weighting schemes achieve ex-post the goals that were established ex-ante, despite the presence of estimation error. That is, the EW portfolio has the maximum effective number of constituents and no turnover by construction. The ERC has the maximum effective number of correlated bets, with the IV following it closely. The MLT-RP-LO, which aims at exploiting the uncorrelated factors of the constituents, displays the highest values for the ENUB and the MDV portfolio reports the highest diversification ratio. Again, depending on the portfolio risk framework, even when considering just 25 factors, optimal risk-minimization allocations can lead to different investment outcomes.

Table 4 shows that there is a substantial difference between the GMV-LO and GMV-N portfolio, displaying lower values of the ENC and the ENCB for the former. The difference between the two GMV portfolios stems from the two constraints that were added to the optimization. Naturally, the GMV tries to equalize the marginal risk contribution of each asset to the overall volatility, thus being heavily exposed to some assets. Imposing a long only constraint, as in the in GMV-LO, reduces the extreme positions and better controls for the presence of estimation error, as pointed out by Jagannathan and Ma (2003). In this set-up, the long only constraint works as a 1-norm controlling extreme weights, short selling and promoting sparse solutions (see Fan et al. 2012). Consequently, the GMV-LO shrinks irrelevant factors towards zero, resulting in low values for the ENC and ENCB. On the other hand, when considering the GMV-N portfolio, the non-linear constraint increases the number of constituents. As a result, the GMV-N is invested in more risk factors than the GMV-LO. As Table 2 suggests, the additional assets fail to push the return upwards, and merely increase the portfolio's volatility, letting the GMV-N underperform. The differences among the portfolios are also confirmed, when examining at the CEQ. The CEQ is the risk-free return that makes an investor indifferent to the risky portfolio strategy. We observe that the GMV-LO exhibits the highest CEQ for both window sizes, whereas the difference to the GMV-N decreases with a longer estimation window.

Considering the risk parity approaches, we have seen before that the IV has the highest variance. This is not surprising, as it disregards the correlation structure and forms the constituents weights, only by considering their volatility. The diversification benefit stemming from less than perfectly correlated assets is thus not exploited. As a result, the IV is fully invested in all factors and ranks second after the EW Portfolio, with regard to the Diversification Ratio (DR).

Finally, the ERC strategy may overweight stocks with lower variance and correlation, but assigns non-zero weights to more portfolio constituents than, for example, the GMV-LO, leading to higher ENC, ENUB, and ENCB. This in turn drives up the volatility, and increases the portfolio return. Furthermore, the EW, IV, and ERC report the lowest Diversification Ratio. The reason for this is the fact that the DR takes into account the weight distribution and the correlation properties between the constituents. For example, portfolios that are concentrated in assets with high correlation tend to have a high volatility, relative to the weighted average of constituents volatilities. As a result, the DR for these portfolios is typically lower. Finally, the MLT-RP-LO, MDC-N, and MDV-N portfolios all perform roughly equal to each other.

Figure 2 complements the information in Table 4, by displaying the portfolio weights decomposition of the 5 risk factors groups for each of the eight strategies over time. While risk is the primary concern for our investor, a high turnover leads to additional costs, which can rendes a promising strategy unprofitable. The area plots in Fig. 2 show the total allocation to the factors (e.g. size), summed over the individual investments in each of the five regions. Not surprisingly, the EW, ERC, and IV allocate roughly 20% continuously to each of the five risk factor groups, resulting in the lowest turnover values among all portfolios.

The two GMV portfolios show substantial differences with the GMV-LO exhibiting dramatic fluctuations in its weight composition over time, leading to the highest turnover among all portfolios. On the other hand, the GMV-N only changes its composition slightly. The stronger variation of the GMV-LO as opposed to the GMV-N, can be explained by the additional 2-norm-constraint, which makes the latter less sensitive to changes in the covariance matrix. Still, these results should be taken with caution. Although an investor should prefer a lower turnover, as the resulting trading cost would otherwise reduce the promised return, the GMV-N ranks among the worst performing portfolios. As such, active management of the portfolio is desirable, being able to move away from underperforming factors. Indeed, we can

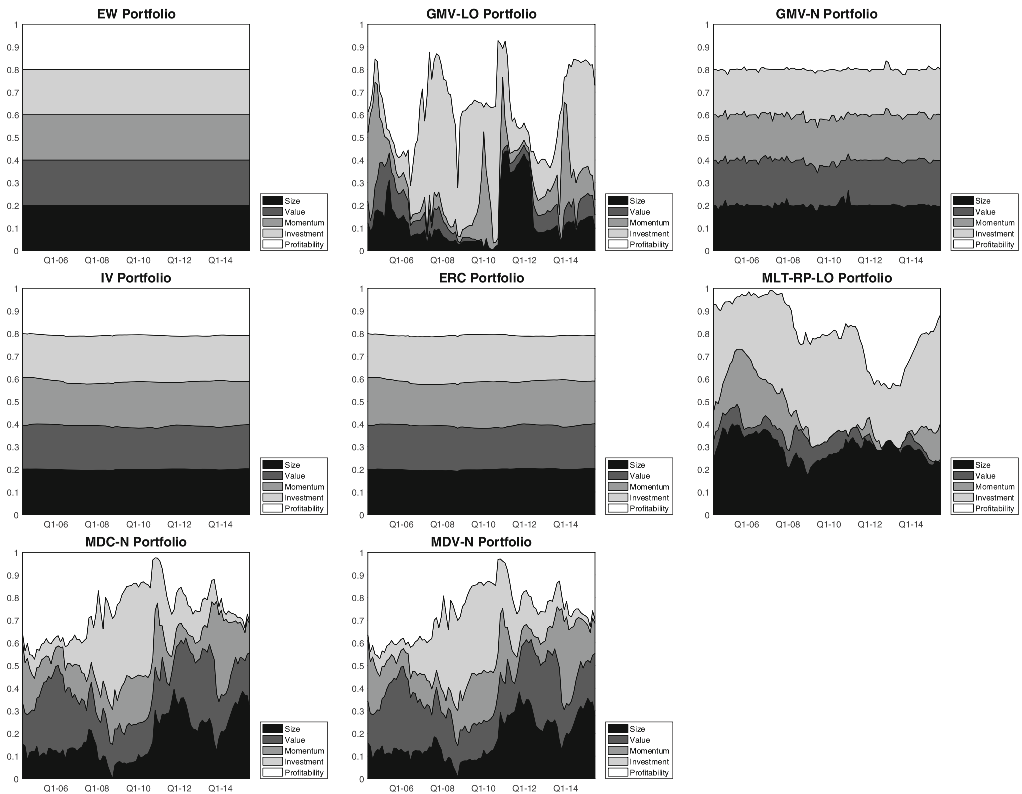

Fig. 2 Weight decomposition of allocation strategies by risk factors. The figure plots the weight decomposition over time for the eight alternative risk minimization allocation schemes. All values representing the investment period from May 2004 to June 2015, considering a rolling window approach of $WS = 500$. The risk-minimization strategies comprise the equal weighting portfolio (EW), the global minimum variance long only portfolio (GMV-LO), the global minimum variance under norm constraints portfolio (GMV-N), the inverse volatility portfolio (IV), the equal risk contribution portfolio (ERC), the minimum linear torsion-risk parity-long only portfolio (MLT-RP-LO), the maximum decorrelation under norm constraints portfolio (MDC-N) and the maximum diversification under norm constraints portfolio (MDV-N)

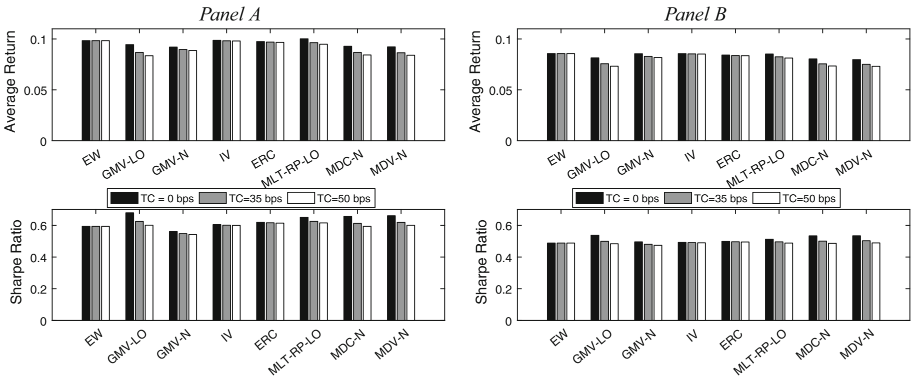

Fig. 3 The figure shows the out-of-sample average annualized return, considering 252 days per year, and the Sharpe ratio, for the eight risk minimization strategies, in the three transaction cost regimes. Transaction costs are assumed to be linear, the same for buying and selling securities and are applied as a multiple of the turnover. We consider three different values of transaction cost: 0, 35 and 50 bps. *Panel A* (*B*) shows the results for a window size of 500 (750) observations. The risk-minimization strategies comprise the equal weighting portfolio (EW), the global minimum variance long only portfolio (GMV-LO), the global minimum variance under norm constraints portfolio (GMV-N), the inverse volatility portfolio (IV), the equal risk contribution portfolio (ERC), the minimum linear torsion-risk parity-long only portfolio (MLT-RP-LO), the maximum decorrelation under norm constraints portfolio (MDC-N) and the maximum diversification under norm constraints portfolio (MDV-N)

observe that the GMV-LO tilts towards the Investment Factor during the crisis period, while underweighting Size, Value, and Momentum factors. This combination of factors changes after the crisis, with the size factor gaining more importance. Equal patterns can be observed for the MDC-N and the MDV-N portfolios, which also underweight Size and Value factors during crisis periods and rank close to the GMV-LO portfolio. Finally, the MLT-RP-LO stands out, by providing a much smoother weight evolution than any of the GMV-LO, GMV-N, MDC-N, or MDV-N portfolios, disregarding the EW, ERC and IV portfolios. Among the non-naive strategies, an investor thus encounters reduced costs, when implementing the MLT-RP-LO allocation.

To evaluate the impact of transaction costs on our risk minimization strategies, we consider a linear transaction cost model, with equal cost for selling and buying securities, applied as a multiple of the turnover. Furthermore, we consider three cost regimes: no, low and high, with cost rates of 0 basis points (bps),[4] 35 bps and 50 bps, respectively. As all of our risk and risk diversification measures remain constant, we only report the changes in the average annual return and the SPR for our eight risk minimization portfolios. To test, whether in each of the cost regimes the STR of the GMV-LO is statistically different from the SPR of the other strategies, we again follow Ledoit and Wolf (2008) and perform the significance test for the SPR. Also here, our results show that none of the strategies are significantly different from the GMV-LO.[5] Figure 3 plots the evolution of the two measures for the three different cost environments for a window size of 500 and 750, respectively. Given the high turnover values of the GMV-LO, we can observe that it has the highest drop in both the return and the SPR, when switching between the no, low and high transaction cost regimes. For costs of 35 bps, our results confirm our previous analysis and still point towards the GMV-LO as the most

[4] 1 bps = 0.01% = 0.0001.

[5] Detailed results for the different cost regimes are available from the authors upon request.

successful strategy. overall, this portfolio is closely followed by the MLT-RP-LO. Indeed, when moving towards the high transaction cost regime, the GMV-LO starts to underperform, while the MLT-RP-LO stands out. This results is not surprising. The higher transaction costs, reduce the return of the GMV-LO portfolio, as it reports the highest turnover, among all portfolios. Consequently, it will be harder to maintain a comparable SPR, when compared to the other strategies. Overall, the ranking of the portfolios, based on the risk measures does not change. The GMV-LO still reports the lowest volatility and superior extreme- and tail risk measures. An investor seeking minimum risk while disregarding returns and costs, is thus still best advised in following this strategy.

5 Conclusion

Traditional asset based investment strategies may lead to portfolios that seem to be well-diversified, but are in fact highly concentrated with regard to certain risk factors. This paper starts by acknowledging the existence and persistence of five risk factors: size, value, momentum, high profitability, and low investment. We consider their evolution for five non-overlapping regions: the US, the UK, Japan, Developed Europe excl. UK, and Asia excl. Japan, and analyse their performance and risk characteristics on a standalone basis and by considering eight alternative state-of-the-art risk-minimization schemes.

Our results confirm that all factors yield positive premia, in exchange for risk, which can lead to considerable underperformance, especially during crisis periods. Our conditional performance analysis shows that no factor consistently performs worse during bad market states. The best-performing regions are the US and Asia excluding Japan. Furthermore, Japan shows the lowest return, while providing the lowest volatility. Moreover, combining risk factors across regions on a portfolio level, enables the investor to exploit diversification benefits, which results in a lower variance, and improved extreme- and tail-risk measures, relative to individual factor investments. Comparing across the eight risk minimization strategies, we show that no portfolio consistently dominates the others with regard to every risk or performance metric. Moreover, the resulting SPRs are not significantly different from each other. Still, an investor might be able to select her preferred strategy based on one individual metric. For example, considering an investor, who wants to achieve minimum risk, the GMV-LO portfolio stands out, showing marginal improvements over the other portfolios. Nevertheless, the strategy is prone to a high turnover, causing the investor to face additional costs.

Our research can be extended in a number of ways. Given that volatility is a key input for almost all risk-minimization weighting schemes and that it penalizes upside risk as much as downside risk, one could argue that a better input instead is a downside risk measure, such as semi-volatility, VaR or CVaR, as recent research by Martellini et al. (2014) proposes. Furthermore, we can expand our study by including methods, that consider Bayesian diffusion priors, which follow a Black-Litterman portfolio approach, or considering higher moments of the distribution, like skewness and kurtosis (see Kourtis et al. 2012; Gulpinar and Pachamanova 2013; Bessler and Wolff 2015; Bessler et al. 2017). Finally, all portfolios have high MDD during the 2008 financial crisis. While diversification is the appropriate tool to efficiently extract long-term risk premia, it is not reliable when it comes to protecting short-term wealth. For this purpose, investors could add a hedging overlay to their portfolios, by buying cheap far out-of-the money put options that provide protection against this kind of tail risk events or rather prefer investing in fewer factors with the lowest riskiness. Further research is then needed to support such insights.

Appendix 1: Risk minimization allocation strategies

Equally weighted portfolio

One of the thoughest benchmarks to beat is the equally weighted (EW) portfolio, which achieves a naïve form of diversification, by investing an equal amount of wealth in each of the N assets, such that:

$$w_i = \frac{1}{N}, \tag{1}$$

where w_i is the weight allocated to the ith-asset, for $i = 1, \ldots, N$.

The EW portfolio is the least concentrated portfolio in terms of weights, having the most evenly spread out dollar amount invested in each asset. It distinguishes itself from other diversification schemes, by being the portfolio with maximum effective number (N) of constituents (ENC).[6] However, the EW portfolio ignores the risk of the assets, either in isolation (volatilities) or when combined into a portfolio (covariances). The different constituents' contributions to portfolio risk may be quite different from the dollar contribution. As a result, an EW portfolio of two assets with very different levels of volatility would have an unbalanced distribution of risk contributions, with most of the risk coming from the asset with largest volatility.

The EW is the best choice when one believes that it is impossible to predict assets volatilities, the individual return parameters of portfolio constituents, and their correlations. It coincides with the efficient portfolio in the mean-variance framework if the expected returns and volatilities of stocks are assumed to be equal and the correlation is uniform. Interestingly, DeMiguel et al. (2009b) show that this naive diversification can lead to better out-of-sample performance, in terms of SPR, certainty-equivalent return and turnover, than optimized portfolios which are generally less efficient, due to estimation errors. On the other hand, DeMiguel et al. (2009a) show that also norm-constrained portfolios can be better alternatives than EW strategies.

Global minimum variance portfolio

Disregarding the mean as an input parameter in the framework of Markowitz (1952) and restricting the weights to be larger than zero, the well-known GMV-LO portfolio can be computed, by solving the following optimization problem:

$$\min_{\boldsymbol{w}\in\mathbb{R}^N} \sigma_p^2 = \boldsymbol{w}'\Sigma\boldsymbol{w} \qquad s.t. \sum_{i=1}^{N} w_i = 1,\ w_i \geq 0\ \forall\ i \in \{1, \ldots, N\}, \tag{2}$$

where σ_p^2 is the variance of portfolio p, $\boldsymbol{w}$ is the $N \times 1$-weight vector of asset weights, and Σ is the $N \times N$-covariance matrix.

Here, we focus on the GMV portfolio under long-only constraints (GMV-LO) and also on the GMV under flexible norm constraints (GMV-N). Fan et al. (2012) show that the long-only constraint works as an ℓ_1-norm, shrinking the portfolio weights towards zero and promoting sparse solutions. In the absence of any constraint, most of the risk reduction, accomplished by the GMV portfolio, comes from concentrating in assets with low volatility, as opposed to exploiting the correlation effects across assets. Adding, to the GMV allocation, flexible concentration constraints on the 2-norm $||\cdot||_2$, instead of rigid upper and lower bounds on

[6] Please refer to Sect. 4.3 for a detailed explanation of the ENC.

individual stock weights, allow for a better use of the correlation structure and improved out-of-sample risk and return properties.

For constructing the GMV-N, we add to (2), the nonlinear constraint that the number of active positions have to be at least one-third of the number of stocks in the portfolio $(N/3)$:

$$||w||_2^2 = \sum_{i=1}^{N} w_i^2 \leq \frac{3}{N} \tag{3}$$

Practically, both constraints set a lower bound on the effective number of constituents. The norm-constraint (3), aims at tackling the problem of concentration in low volatility stocks that is specific to GMV allocation (Clarke et al. 2011).

Equal risk contribution portfolio

The Equal Risk Contribution (ERC) portfolio aims to equalize the marginal risk contributions of different assets in the portfolio. In the case of the ERC approach, constituents that either have lower volatilities, lower correlation with the other assets, or both, are assigned a larger weight in the portfolio. Given that portfolio variance can be decomposed as:

$$\sigma_p^2 = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \sigma_{ij} = \sum_{i=1}^{N} w_i \sum_{j=1}^{N} w_j \sigma_{ij} \tag{4}$$

the marginal contribution to the portfolio risk for asset i is given as:

$$c_i^{var} = w_i \sum_{j=1}^{N} w_j \sigma_{ij} = w_i \ (\Sigma \boldsymbol{w})_i \ \text{ with } \ \sum_{i=1}^{N} c_i^{var} = \sigma_p^2 \tag{5}$$

where $(\Sigma \boldsymbol{w})_i$ denotes the ith row of the product of Σ and $\boldsymbol{w}$.

The ERC portfolio aims to equalize risk contributions from all assets in the portfolio, but does not have an analytical solution. The reason for this is that the weights are endogenous in determining the risk contribution of each asset in the portfolio. However, the solution can be obtained numerically as:

$$\begin{aligned} &\min_{\boldsymbol{w} \in \mathbb{R}^N} \sum_{i=1}^{N} \left(\frac{w_i \ (\Sigma \boldsymbol{w})_i}{\sigma_p^2} - \frac{1}{N} \right)^2 \\ &s.t. \ \sum_{i=1}^{N} w_i = 1, \ \ 0 \leq w_i \leq 1 \ \ \forall \, i \in \{1, 2, \ldots, N\} \end{aligned} \tag{6}$$

The ERC portfolio is less sensitive to small changes in the covariance matrix than the GMV portfolio. Maillard et al. (2010) show that the ERC portfolio coincides with the optimal GMV portfolio, when the correlation and the SPR are equal among all assets. It may be viewed as a type of variance-minimizing portfolio, subject to a constraint of sufficient diversification in terms of component weights and with a volatility between that of the GMV and the EW portfolio.

The drawbacks of this methodology are the fact that the “marginal” contribution to risk involves an arbitrary attribution of the correlated components, i.e. the contribution of asset i to the portfolio variance depends on the covariance of asset i with the other assets included in the portfolio. Furthermore, it overlooks the fact that well-diversified portfolios, from the point of equal dollar or equal risk contribution of each constituent, may actually be heavily exposed to a small number of independent factors. Hence, they are not so diversified with regard to independent factor exposures. To circumvent these drawbacks the investor can consider a MLT-RP methodology, as presented below.

Minimum linear torsion: risk parity portfolio

The Minimum Linear Torsion-Risk Parity (MLT-RP) portfolio equalises the contribution of independent risk factors, rather than the contribution of individual constituents (Meucci et al. 2014). This methodology involves, as an additional step, the transformation of correlated asset returns into uncorrelated risk factors that are the closest to the initial assets in terms of tracking error. It is based on the diversification measure proposed by Meucci et al. (2014), namely the entropy-based measure of the effective number of uncorrelated minimum-torsion bets (ENUB).[7] Central to this methodology is that the portfolio return can be expressed as a combination of uncorrelated factors obtained through the Minimum Linear Torsion methodology. The advantage of introducing these uncorrelated factors is that the portfolio variance can be decomposed as a sum of the contributors of each factor's variance:

$$\boldsymbol{w}'\Sigma\boldsymbol{w} = \boldsymbol{w}'(T')^{-1}\Sigma_F T^{-1}\boldsymbol{w} \tag{7}$$

$$= \boldsymbol{w}_F'\Sigma_F\boldsymbol{w}_F \tag{8}$$

$$= \sum_{k=1}^{N}(\sigma_{F_k} w_{F_k})^2, \tag{9}$$

where T is the decorrelated Torsion matrix, and w_{F_k} and σ_{F_k} is the weight and the risk corresponding to the minimum-torsion independent factor k, respectively.

It follows that the relative contribution to total risk from each uncorrelated bet is:

$$p_k \equiv \frac{v\{w_{F_k} r_{F_k}\}}{v\{r_p\}}, \quad \forall\ k = 1, \ldots, N, \quad with \sum_{k=1}^{K} p_k = 1, \tag{10}$$

where $v\{w_{F_k} r_{F_k}\}$ represents the variance of the minimum torsion independent factor k, and $v\{r_p\}$ the variance of the portfolio returns.

The MLT-RP-LO portfolio aims at equalizing the risk contributions from all uncorrelated bets in the portfolio. Similarly to the ERC approach, the MLT-RP-LO portfolio does not have an analytical solution because portfolio weights are endogenous in determining the risk contribution of each minimum-torsion independent factor in the portfolio. However, the solution can be obtained numerically by:

$$\min_{\boldsymbol{w}\in\mathbb{R}^N} \sum_{i=1}^{N}\left(\frac{v\{w_{F_k} r_{F_k}\}}{v\{r_p\}} - \frac{1}{N}\right)^2, \ s.t. \sum_{i=1}^{N} w_i = 1,\ 0 \le w_i \le 1\ \forall\, i\ \in\ \{1, 2, \ldots, N\} \tag{11}$$

See Meucci et al. (2014) for a full description.

Inverse volatility portfolio

Maillard et al. (2010) show that under the assumption of uniform cross-correlations between the constituents, the risk parity portfolio weights are proportional to the inverse of the stocks' individual volatilities. This is the Inverse Volatility (IV) portfolio.

[7] See Sect. 4.3 for a detailed description of the ENUB.

Assuming equal pairwise correlations $\rho_{ij} = \rho$ for all $i \neq j$, the risk contribution of asset i to portfolio variance can be written as:

$$\frac{c_i^{var}(w)}{\sigma_p^2} = \frac{w_i^2\sigma_i^2 + \rho\sum_{i\neq j} w_i w_j \sigma_i \sigma_j}{\sigma_p^2} = \frac{w_i\sigma_i((1-\rho)w_i\sigma_i + \rho\sum_j w_j\sigma_j}{\sigma_p^2} \tag{12}$$

In order to have:

$$\frac{c_i^{var}}{\sigma_p^2} = \frac{c_j^{var}}{\sigma_p^2}\ \forall\, i \neq j,\ s.t. \sum_{i=1}^{N} w_i = 1, \tag{13}$$

Maillard et al. (2010) show that the solution to (13) is given by:

$$w_i = \frac{\frac{1}{\sigma_i}}{\sum_{i=1}^{N}\frac{1}{\sigma_i^2}}, \forall\, i = 1, \ldots, N, \tag{14}$$

where σ_i represents the volatility of asset i.

In other words, if one trusts the volatility estimates, and assuming constant correlation, thus disregarding estimation errors in the correlation matrix, an alternative approach to target risk parity is to weight assets by the inverse of their volatility.

A key advantage of this approach is its analytical simplicity and its intuitive appeal: the higher (lower) the volatility of a component, the lower (higher) its weight in the IV portfolio. The disadvantages of the IV portfolio include that investors cannot exploit the differences between cross-correlations in the weighting of the stocks. Maillard et al. (2008) provide evidence that the weights of the ERC portfolio may differ significantly from those of the IV portfolio, especially when both negative and positive correlations exist. The authors report that the differences seem to increase with the number of constituents. However, the two portfolios' weights differ less, when there are only positive coefficients in the correlation matrix, which is generally the case for long-only portfolios exposed solely to equity risk.

Maximum diversification portfolio

The Maximum Diversification (MDV) portfolio, formally introduced by Choueifaty and Coignard (2008), aims at maximizing the diversification ratio, defined as the ratio of the weighted average asset volatility to overall portfolio volatility:

$$\max_{\boldsymbol{w}\in\mathbb{R}^N} DR(\boldsymbol{w}) = \frac{\sum_{i=1}^{N} w_i\sigma_i}{\sqrt{\boldsymbol{w}'\Sigma\boldsymbol{w}}},\quad s.t.\ \sum_{i=1}^{N} w_i = 1,\ \ \sum_{i=1}^{N} w_i^2 \leq \frac{3}{N},\ \ w_i \geq 0, \forall\, i \in \{1, \ldots, N\}, \tag{15}$$

where $\boldsymbol{w}$ is the weights vector and $DR(\boldsymbol{w})$ is the Diversification Ratio.

The DR is greater than 1 for portfolios made up of assets that are not perfectly correlated, as the relatively high volatility of component assets results in overall lower portfolio volatility through the risk reduction effect of correlations. The MDV portfolio coincides with the tangency portfolio, if the SPR for all assets are equal, or, in other words, the excess returns of assets are proportional to their volatilities.

Maximum decorrelation portfolio

Another portfolio that aims at minimizing portfolio volatility, but better exploits the information provided by the mutual correlation coefficients is the Maximum Decorrelation (MDC) portfolio. The idea behind the MDC portfolio is to combine stocks, as to exploit the risk reduction effect stemming from low correlations, rather than reducing risk by concentrating in low volatility constituents. The MDC portfolio aims at minimising the portfolio volatility under the assumption that individual volatilities are identical, that is:

$$\min_{\boldsymbol{w}\in\mathbb{R}^N} C(\boldsymbol{w}) = \boldsymbol{w}'\Omega\boldsymbol{w}, \quad s.t. \quad \sum_{i=1}^{N} w_i = 1, \quad \sum_{i=1}^{N} w_i^2 \leq \frac{3}{N}, \quad w_i \geq 0, \forall\, i \in \{1, \ldots, N\}, \tag{16}$$

where $\boldsymbol{w}$ is the weight vector and Ω is the correlation matrix.

The MDC portfolio uses only the information provided by the correlation structure. Still, the risk reduction provided by the GMV framework can potentially also be attained by the MDC portfolio, which avoids concentration by ignoring differences in individual volatilities. It is worth noting that this approach implies no explicit control on the total portfolio risk that results from the volatility of each constituent.

Appendix 2: Risk factors

Size premium The size premium captures the effect that small stocks (stocks with lower market capitalisation) on average outperform large stocks. Fama and French (2007) found evidence that the size premium is the result of small stocks earning extreme positive returns and becoming bigger stocks. Souza (2014) analyses the size premium over a long term period of 85 years and finds that the premium is positive most of the time, and statistically significant especially during time periods when risk premia are high (periods with high book-to-market equity of the aggregate stock market).

Value premium[8] The value premium is the well established finding that stocks with high book-to-market ratios (value stocks) yield higher average returns than those with low book-to-market ratios (growth stocks). In other words, value investments favour stocks that have a low price relative to fundamental firm characteristics. Economic explanations for this finding includes among others, that value firms are riskier than growth firms during turbulent periods, when the price of risk is high (Zhang 2005), or that firms with a high book-to-market ratio can benefit more from positive shocks in the real economy than low book-to-market firms, because they have excess capital (Cooper 2006).

Momentum premium The pioneering research of Jegadeesh and Titman (1993) suggests that if stock prices overreact or underreact to information, then there exist profitable trading strategies that select stocks based on their past returns. The authors show that most of the returns of the zero-cost portfolios (i.e. long winners minus short losers) are positive. Rouwenhorst (1998) supports these results, by using an internationally diversified relative strength portfolio, which invests in medium-term winners and sells past medium-term losers, showing that the momentum in returns can be found in all 12 countries of the sample. The economic rationale for the momentum premium was advocated by Chordia and Shivakumar (2002), who argue that the profit of the momentum strategies is explained by common macroeconomic variables that are related to the business cycle, while Liu and Zhang (2008) found that

[8] It might be argued, following Fama and French (2006), that the value factor is redudant, given that one already considers the market, the size, the investment and the profitability premium. However, in the presence of the momentum factor, Asness et al. (2015) show that adding a value tilt does enhance the risk diversification of a factor portfolio.

winners have temporarily higher loadings on the growth rate of industrial production than recent losers, and that this exposure explains more than half of the momentum profits.

High profitability and low investment premium Profitability is typically proxied by the Return on Equity (ROE), defined as net income divided by shareholders' equity. Investment is typically characterized by asset growth, defined as the change in book value of assets over previous years. Novy-Marx (2013) shows that profitable firms generate higher returns than unprofitable firms. Cooper et al. (2008) show that a firm's asset growth is an important determinant of stock returns. In their analysis, low investment firms (firms with low asset growth rates) generate about 8% annual outperformance over high investment firms. Titman et al. (2004) show a negative relation between investment, which they measure by the growth of capital expenditures, and stock returns in the cross section.

Regarding the economic justification for profitability and investment factors, two studies are commonly stated: (1) Fama and French (2006) derive the relation between book-to-market ratio, expected investment, expected profitability and expected stock returns from the dividend discount model, arguing that (a) controlling for book-to-market and expected growth in book equity, more profitable firms (firms with high earnings relative to book equity) have higher expected return; and (b) controlling for book-to-market and profitability, firms with higher expected growth in book equity (high reinvestment of earnings) have lower expected returns. Finally, (2) Hou et al. (2015) argue that high profitability and low investments are driven by a high discount rate. Said differently, if the discount rate was not high enough to offset the high profitability—i.e. providing investments with pos. NPV given the high discount rate—the firm would face many investment opportunities with positive NPV and thus invest more by accepting less profitable investments.

Appendix 3: Risk- and return measures

In the following section, we will explain our used risk and return metrics in more detail. Where applicable, we give further comments beside the plain mathematical formulation. Otherwise, we restrict ourselves to a brief explanation and refer for more details to the given references. For all measures, assume that we have N factors and T out of sample observations. Now, let $\boldsymbol{w}_t = [w_{1,t}, \ldots, w_{N,t}]'$, be the weight vector of the factor portfolio for one of the eight risk minimization allocation strategies at rebalancing period t, and where $w_{n,t}$ is, respectively, the idiosyncratic weight for factor k at time t. Furthermore, let $\boldsymbol{r}_t = [r_1, \ldots, r_N]'$, be the return vector of the N risk factors at time t, with $r_{n,t}$ being the idiosyncratic return of factor n at time t. Then the out of sample portfolio return time series is given as $\boldsymbol{R} = [R_1, \ldots, R_T]'$, where $R_t = \boldsymbol{w}_t' \boldsymbol{r}_t$ is the portfolio return at time t. Given $\boldsymbol{R}$, we compute the following risk and return metrics for each risk minimization portfolio allocation strategy:

Average return

$$\mu_p = \frac{1}{T} \sum_{t=1}^{T} R_t \tag{17}$$

Volatility

$$\sigma_p = \sqrt{\frac{\sum_{t=1}^{T} (R_t - \mu_p)^2}{T - 2}} \tag{18}$$

Loss standard deviation

$$LSD = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T} \min(0, R_t)^2} \tag{19}$$

Active positions

$$AP = \frac{1}{T}\sum_{t=1}^{T} AP_t \quad \text{with} \quad AP_t = \#\{\boldsymbol{w}_t | w_{n,t} \neq 0\} \tag{20}$$

Turnover

$$TO = \frac{1}{T}\sum_{t=2}^{T}\sum_{i=1}^{N} |w_{i,t} - w_{i,t-1}| \tag{21}$$

Value at risk The historical Value at Risk (VaR 5%) is based on historical returns over the entire sample and is defined as the maximum loss an investor can occur in a certain time period. It is defined as the α quantile of the return distribution. Formally given as:

$$VaR_\alpha(\boldsymbol{R}) = \inf\{l : Pr\{-\boldsymbol{R} < l\} \geq \alpha\} \tag{22}$$

Expected shortfall Closely related to the VaR is the Expected Shortfall (ES). The historical ES at 5% calculates the average losses given the portfolio loses which are more than the historical VaR at 5%.

$$ES_\alpha(\boldsymbol{R}) = \frac{1}{1-\alpha}\int_\alpha^1 Var_u(\boldsymbol{R})\, du \tag{23}$$

Maximum drawdown The maximum drawdown (MDD) is the maximum loss from peak to through that an investor incurs computed over the entire sample period. Formally it is defined as:

$$MDD(\boldsymbol{R}) = \frac{(P-L)}{P}, \tag{24}$$

where P is the peak value before the largest drop and L is the lowest value before the new high has been established.

Maximum drawdown period and recovery period While the MDD Period indicates the years over which the investor incurred losses between the peak and valley of the maximum drawdown, the Recovery Period after MDD, represents the years it takes the investor to bring the portfolio value back above the peak of the maximum drawdown.

Certainty equivalent The certainty equivalent (CEQ) is defined as the risk-free-rate that makes a risk averse investor indifferent to a risky gamble. We compute the CEQ as:

$$CEQ = \mu_p - \frac{\gamma}{2} \times \sigma_p^2, \tag{25}$$

for which we follow DeMiguel et al. (2009a) and set γ, the risk aversion parameter, to 5.

Effective number of constituents (ENC) The ENC is a simplistic indicator of the nominal holdings of the portfolio. More formally, it is defined as the exponential of the entropy, given the distribution of the portfolio weight vector. Given the T different portfolio observations, we compute the average ENC as:

$$ENC = \frac{1}{T}\sum_{t=1}^{T} ENC(\boldsymbol{w}_t)_t \quad \text{where} \quad ENC(\boldsymbol{w}_t)_t \equiv exp(-\sum_{i=1}^{N} w_{i,t}\, ln(w_{i,t})) \tag{26}$$

The ENC has a minimum value of one when the portfolio is fully concentrated in one constituent, and a maximum value of N, when the portfolio is equally weighted and in which N represents the size of the investment universe. Although intuitively appealing, it disregards the volatility and correlation structure of the chosen universe. As a result, an ENC that is close to N indicates a diversified portfolio, which, however, can be very concentrated in terms of risk. Deguest et al. (2013)

Effective number of correlated bets (ENCB) The ENCB builds on the limitations of the ENC and incorporates information on the volatility and correlation structure of the constituents. The ENCB is lower, given a high volatility asset, which also has a high positive correlation with other constituents. Formally, it is represented by the entropy of the distribution of the correlated constituents' contribution to the total variance. To compute the ENCB, recapture that the variance of the portfolio is given as:

$$\sigma_{p,t}^2 = \boldsymbol{w_t}'\boldsymbol{\Sigma_t}\boldsymbol{w_t} = \sum_{i=1}^{N} w_{i,t}|\boldsymbol{\Sigma_t}\boldsymbol{w}_t|_i \tag{27}$$

where $|X|_i$ denotes the i^{th} element of matrix X. Now, the contribution of each constituent to the overall variance can be expressed as:

$$q_{i,t} = \frac{w_{i,t}|\boldsymbol{\Sigma_t}\boldsymbol{w}_t|_i}{\boldsymbol{w_t}'\boldsymbol{\Sigma}_t\boldsymbol{w}_t} \tag{28}$$

The ENCB is now defined as the entropy of the distribution of the effective number of correlated factors contribution to the total variance:

$$ENCB = \sum_{t=1}^{T} ENCB(\boldsymbol{w}_t)_t \text{ with } ENCB(\boldsymbol{w}_t)_t = exp(-\sum_{i=1}^{N} q_{i,t}\ ln(q_{i,t})) \tag{29}$$

The ENCB has a minimum value of one when the risk of the portfolio is fully concentrated in one asset and a maximum value of N, when the risk is evenly spread among the assets. Roncalli (2013)

Effective number of uncorrelated bets (ENUB) Closely related to the ENCB is the ENUB, which is obtained by looking at the uncorrelated factors using the Minimum Linear Torsion methodology. In detail, Meucci et al. (2014) propose to define the portfolio return not as the weighted average of its constituents, but rather as the combination of uncorrelated factors, obtained through the Minimum Linear Torsion methodology. The ENUB is then defined as the exponential of the entropy of the distribution of the uncorrelated minimum torsion factors' that contribute to the total variance. Formally, after having obtained the $N \times N$ torsion factor matrix $\boldsymbol{H}$, the contribution from the truly separate sources of risk is given as:

$$p_{w,t} = \frac{(\boldsymbol{H'}_t^{-1}\boldsymbol{w}_t) \circ (\boldsymbol{H}_t\boldsymbol{\Sigma}_t\boldsymbol{w}_t)}{\boldsymbol{w}_t'\boldsymbol{\Sigma}_t\boldsymbol{w}_t} \tag{30}$$

where $\circ$ is the term by term product. The ENUB is the entropy-based measure given as:

$$ENUB(\boldsymbol{w}_t) = exp(-\sum_{i=1}^{N} p_{i,t}\ ln(p_{i,t})) \tag{31}$$

Like the ENCB, the ENUB has a minimum value of 1 when the risk of the portfolio is fully concentrated in one asset and a maximum value of N, when the risk is evenly spread amongst the assets.

Diversification ratio The Diversification Ratio (DR) was introduced by Choueifaty et al. (2011), and measures the ratio of the weighted variance of the portfolio constituents to the overall portfolio's variance. It is thus a natural indicator for the diversification exploited by investing in the individual constituents. We report the average DR across all rebalancing periods as:

$$DR = \frac{1}{T}\sum_{t=1}^{T} DR_t(\boldsymbol{w}_t) \quad \text{with} \quad DR_t(\boldsymbol{w}_t) = \frac{\sum_{i=1}^{N} w_{i,t}\boldsymbol{\Sigma}_{i,t}}{\sqrt{\boldsymbol{w}_t'\boldsymbol{\Sigma}_t\boldsymbol{w}_t}} \tag{32}$$

The DR is greater than one for portfolios composed of assets that are not perfectly correlated, as the relatively high volatility of component assets results in overall lower portfolio volatility through the risk reduction effect of correlations.

Appendix 4: Market conditions

Bull/bear markets Stock market returns are ranked in descending order. Bear market comprises the months with market returns below the median monthly return, computed considering the whole period, and Bull market comprises the rest. To represent the market returns, the following equity indices were used: S&P 500 for the US universe, FTSE 100 for the UK universe, Nikkei 225 for the Japanese universe, MSCI AC Asia ex-Japan for Asia ex-Japan universe, and STOXX Europe 600 for the European universe.

Low/high volatility market High and low volatility conditions are differentiated by using the one month realized volatility of the equity indices in each region (S&P 500, FTSE 100, Nikkei 225, MSCI AC Asia ex-Japan, and STOXX Europe 600). The monthly volatility is ranked in descending order. Low Volatility market comprises the months with realized volatility below the median monthly volatility and High Volatility market comprises the rest.

High/low interest rate regime High and low interest rate conditions are differentiated by using the changes in term spread. The term spread is the difference between the yields of the 10-year and 2-year government bonds in each region (for Developed Europe ex-UK and Asia ex-Japan, the German Bonds and Chinese Government Bonds, respectively were used as proxies; for US, UK and Japan, the bonds issued by the national governments were used). A positive change in the term spread means high interest rates and a negative change means low interest rates. High interest rate market conditions are associated with an expansion period as the monetary policy rate is usually set high in order to prevent the overheating of the economy. Conversely, low interest rate market conditions are associated with a recession period as the monetary policy rate is usually set low in order to encourage consumption and investment.

Low/high credit risk Low and high credit risk conditions are differentiated using the iTraxx indexes, which track the Credit Default Swap (CDS) spread on the largest investment-grade companies from each region. A widening (narrowing) of the CDS spread is associated with deteriorating (improving) credit conditions. Consequently, a positive change in the CDS spread means high credit risk and a negative change in the credit spread means low credit risk.

Positive/negative economic surprises Positive and negative economic surprises periods are differentiated by using the Citi Economic Surprise Index for each region. This indicator measures data surprises relative to market expectations. A positive reading means that data releases have been stronger than expected and a negative reading means that data releases have been worse than expected.

Table 5 Conditional performance comparison of factor-tilted indices

	Market		Volatility		Credit		Interest		Surprises		Conditions	
	Bull	Bear	Low	High	Low	High	High	Low	Positive	Negative	Good	Bad
US												
Size												
Return	**4.04%**	**−1.82%**	**2.21%**	**0.02%**	**3.09%**	**−0.37%**	1.14%	1.09%	1.03%	1.21%	1.57%	0.16%
Volatility	4.04%	5.44%	**2.89%**	**6.60%**	3.56%	3.81%	4.84%	4.66%	4.45%	5.07%	**3.89%**	**6.53%**
RTR	**1.15**	**−0.20**	**0.81**	**0.14**	**0.98**	**−0.03**	0.49	0.46	0.49	0.46	0.59	0.24
Value												
Return	**3.95%**	**−1.77%**	**2.08%**	**0.10%**	**2.91%**	**−0.11%**	1.17%	1.03%	1.08%	1.11%	1.49%	0.25%
Volatility	4.03%	5.45%	**2.78%**	**6.70%**	3.52%	3.70%	4.93%	4.57%	4.48%	5.05%	**3.81%**	**6.69%**
RTR	**1.16**	**−0.19**	**0.83**	**0.14**	**0.93**	**0.06**	0.49	0.47	0.49	0.48	0.59	0.26
Momentum												
Return	**3.82%**	**−1.76%**	**2.28%**	−0.22%	**2.99%**	**−0.44%**	0.91%	1.13%	0.83%	1.26%	1.55%	−0.07%
Volatility	**3.96%**	**5.46%**	**2.97%**	**6.45%**	3.60%	3.93%	4.69%	4.73%	4.43%	5.03%	**4.01%**	**6.17%**
RTR	**1.11**	**−0.18**	**0.82**	**0.10**	**0.96**	**−0.02**	0.46	0.46	0.47	0.45	0.59	0.20
Investment												
Return	**3.88%**	**−1.63%**	**2.15%**	**0.10%**	**2.98%**	**−0.03%**	1.23%	1.04%	1.11%	1.14%	1.53%	0.27%
Volatility	3.74%	5.07%	**2.66%**	**6.15%**	3.33%	3.58%	4.54%	4.29%	4.14%	4.72%	**3.64%**	**6.02%**
RTR	**1.20**	**−0.18**	**0.87**	**0.14**	**1.02**	**0.07**	0.52	0.49	0.52	0.49	0.63	0.26
Profitablitiy												
Return	**3.85%**	**−1.62%**	**2.11%**	**0.11%**	**2.94%**	**−0.18%**	1.30%	0.95%	0.96%	1.29%	1.47%	0.36%
Volatility	3.81%	5.09%	**2.78%**	**6.13%**	3.46%	3.74%	4.53%	4.38%	4.13%	4.83%	**3.72%**	**5.98%**
RTR	**1.13**	**−0.19**	**0.79**	**0.14**	**0.99**	**0.04**	0.52	0.42	0.48	0.46	0.57	0.25
Japan												
Size												
Return	**3.02%**	**−2.05%**	1.20%	−0.24%	1.88%	−1.67%	0.96%	0.14%	0.26%	0.63%	0.94%	−0.22%

Table 5 continued

	Market		Volatility		Credit		Interest		Surprises		Conditions	
	Bull	Bear	Low	High	Low	High	High	Low	Positive	Negative	Good	Bad
Volatility	4.66%	5.59%	**3.88**%	**6.37**%	3.91%	4.43%	5.27%	5.02%	5.24%	5.05%	**4.62**%	**5.92**%
RTR	**0.69**	−**0.32**	0.32	0.05	0.50	−0.38	0.25	0.13	0.23	0.15	0.23	0.10
Value												
Return	**3.30**%	−**2.27**%	1.33%	−0.30%	**2.27**%	−**2.11**%	1.03%	0.14%	0.44%	0.56%	1.07%	−0.35%
Volatility	4.90%	5.79%	**4.06**%	**6.63**%	4.11%	4.64%	5.52%	5.21%	5.50%	5.24%	**4.78**%	**6.22**%
RTR	**0.72**	−**0.37**	0.33	0.02	**0.59**	−**0.44**	0.26	0.11	0.28	0.11	0.26	0.05
Momentum												
Return	**3.21**%	−**2.24**%	1.42%	−0.46%	1.98%	−1.93%	0.92%	0.16%	0.24%	0.64%	1.03%	−0.36%
Volatility	4.85%	5.82%	**4.06**%	**6.61**%	4.28%	4.76%	5.42%	5.28%	5.55%	5.19%	**4.93**%	**5.97**%
RTR	**0.70**	−**0.33**	0.35	0.02	0.50	−0.38	0.24	0.14	0.22	0.15	0.24	0.10
Investment												
Return	**3.21**%	−**2.19**%	1.33%	−0.31%	2.47%	−2.34%	1.09%	0.08%	0.50%	0.51%	1.02%	−0.29%
Volatility	4.81%	5.70%	**4.04**%	**6.48**%	4.11%	4.74%	5.37%	5.17%	5.47%	5.11%	**4.79**%	**5.98**%
RTR	**0.72**	−**0.36**	0.34	0.02	**0.63**	−**0.49**	0.27	0.12	0.28	0.12	0.25	0.08
Profitablitiy												
Return	**3.07**%	−**1.93**%	1.24%	−0.09%	2.25%	−1.41%	1.10%	0.18%	0.17%	0.84%	0.95%	−0.02%
Volatility	4.56%	5.47%	**3.75**%	**6.28**%	3.96%	4.53%	5.19%	4.89%	5.15%	4.93%	**4.48**%	**5.85**%
RTR	**0.71**	−**0.31**	0.33	0.07	0.59	−0.31	0.29	0.13	0.20	0.19	0.23	0.14
UK												
Size												
Return	**3.20**%	−**1.37**%	1.66%	0.17%	–	–	1.20%	0.65%	1.00%	0.81%	1.66%	−0.44%
Volatility	5.53%	5.82%	**3.64**%	**7.71**%	–	–	5.96%	5.42%	5.93%	5.34%	**4.93**%	**7.04**%
RTR	**0.73**	−**0.05**	**0.54**	**0.13**	–	–	0.39	0.29	0.36	0.31	0.47	0.10

Table 5 continued

	Market		Volatility		Credit		Interest		Surprises		Conditions	
	Bull	Bear	Low	High	Low	High	High	Low	Positive	Negative	Good	Bad
Value												
Return	**3.56%**	**−1.93%**	1.71%	−0.07%	–	–	1.13%	0.53%	1.01%	0.57%	1.57%	−0.56%
Volatility	5.73%	6.09%	**3.63%**	**8.19%**	–	–	6.29%	5.56%	6.23%	5.51%	**5.15%**	**7.29%**
RTR	**0.76**	**−0.11**	**0.55**	**0.10**	–	–	0.36	0.29	0.36	0.28	0.46	0.08
Momentum												
Return	**3.46%**	**−1.40%**	2.02%	0.04%	–	–	1.00%	1.06%	1.14%	0.89%	1.80%	−0.38%
Volatility	5.35%	5.76%	**3.64%**	**7.46%**	–	–	5.86%	5.27%	5.75%	5.31%	**4.86%**	**6.82%**
RTR	**0.82**	**−0.07**	**0.62**	**0.13**	–	–	0.35	0.39	0.42	0.32	0.52	0.11
Investment												
Return	**3.30%**	**−1.39%**	1.75%	0.16%	–	–	1.19%	0.74%	1.23%	0.60%	1.63%	−0.28%
Volatility	5.16%	5.55%	**3.42%**	**7.29%**	–	–	5.65%	5.08%	5.60%	5.04%	**4.63%**	**6.67%**
RTR	**0.78**	**−0.05**	**0.58**	**0.15**	–	–	0.40	0.33	0.44	0.27	0.49	0.13
Profitablitiy												
Return	**3.25%**	**−1.17%**	1.70%	0.38%	–	–	1.16%	0.93%	1.26%	0.75%	1.65%	−0.08%
Volatility	5.05%	5.44%	**3.34%**	**7.15%**	–	–	5.62%	4.90%	5.44%	4.99%	**4.47%**	**6.65%**
RTR	**0.79**	**−0.03**	0.59	0.18	–	–	0.38	0.39	0.44	0.31	0.51	0.14
Europe ex-UK												
Size												
Return	**3.96%**	**−2.07%**	**2.49%**	**−0.60%**	**3.27%**	**−1.63%**	0.06%	1.52%	**2.25%**	**−0.44%**	**2.22%**	**−1.30%**
Volatility	**4.65%**	**6.05%**	**3.55%**	**7.15%**	4.75%	4.76%	**6.18%**	**4.81%**	4.81%	5.92%	**4.74%**	**6.43%**
RTR	**1.01**	**−0.19**	**0.82**	**0.00**	**0.87**	**−0.41**	0.36	0.45	**0.67**	**0.14**	**0.64**	**0.01**
Value												
Return	**4.60%**	**−2.69%**	**2.71%**	**−0.80%**	**3.90%**	**−2.41%**	0.09%	1.51%	**2.35%**	**−0.53%**	**2.36%**	**−1.53%**
Volatility	**5.32%**	**6.94%**	**3.94%**	**8.32%**	5.56%	5.62%	7.13%	5.48%	5.40%	6.90%	**5.37%**	**7.47%**
RTR	**1.01**	**−0.22**	**0.81**	**−0.02**	**0.86**	**−0.45**	0.34	0.43	**0.65**	**0.12**	**0.61**	**0.00**

Table 5 continued

	Market		Volatility		Credit		Interest		Surprises		Conditions	
	Bull	Bear	Low	High	Low	High	High	Low	Positive	Negative	Good	Bad
Momentum												
Return	**4.03%**	**−1.91%**	**2.69%**	**−0.58%**	**3.22%**	**−1.27%**	0.19%	1.61%	**2.29%**	**−0.25%**	**2.34%**	**−1.21%**
Volatility	**4.66%**	**6.12%**	**3.65%**	**7.13%**	4.62%	4.74%	6.18%	4.88%	4.91%	5.90%	4.91%	6.24%
RTR	**1.01**	**−0.16**	**0.83**	**0.02**	**0.86**	**−0.29**	0.38	0.45	**0.66**	**0.17**	**0.65**	**0.03**
Investment												
Return	**4.14%**	**−1.99%**	**2.56%**	**−0.42%**	**3.69%**	**−2.00%**	0.35%	1.54%	**2.31%**	**−0.24%**	**2.30%**	**−1.09%**
Volatility	**4.80%**	**6.21%**	**3.67%**	**7.33%**	5.08%	5.18%	6.32%	4.97%	4.91%	6.13%	**4.86%**	**6.64%**
RTR	**0.99**	**−0.16**	**0.81**	**0.02**	**0.91**	**−0.40**	0.36	0.45	**0.66**	**0.15**	**0.63**	**0.03**
Profitablitiy												
Return	**3.81%**	**−1.65%**	**2.39%**	**−0.23%**	**3.06%**	**−1.22%**	0.27%	1.61%	2.22%	−0.13%	**2.24%**	**−0.95%**
Volatility	**4.49%**	**5.91%**	**3.44%**	**6.96%**	4.40%	4.52%	**6.05%**	**4.65%**	4.71%	5.72%	**4.68%**	**6.12%**
RTR	**0.98**	**−0.15**	**0.78**	**0.05**	**0.84**	**−0.33**	0.36	0.45	**0.66**	**0.16**	**0.63**	**0.04**
ASIA ex-Japan												
Size												
Return	**5.34%**	**−2.90%**	**2.72%**	**−0.28%**	**2.65%**	**−1.56%**	1.81%	−0.04%	1.29%	1.10%	1.61%	−1.37%
Volatility	4.11%	5.48%	**3.00%**	**6.59%**	3.38%	3.38%	5.07%	4.95%	4.82%	4.75%	**4.22%**	**8.60%**
RTR	**1.44**	**−0.38**	**0.90**	**0.16**	**0.87**	**−0.46**	0.58	0.27	0.63	0.36	0.61	0.02
Value												
Return	**5.57%**	**−3.09%**	**2.75%**	**−0.27%**	**2.57%**	**−1.93%**	1.76%	−0.08%	1.44%	0.90%	1.61%	−1.23%
Volatility	4.40%	5.76%	3.15%	7.02%	3.52%	3.47%	5.35%	5.21%	5.09%	5.07%	**4.42%**	**9.48%**
RTR	**1.41**	**−0.45**	**0.83**	**0.13**	**0.79**	**−0.62**	0.51	0.23	0.62	0.24	0.55	0.02

Table 5 continued

	Market		Volatility		Credit		Interest		Surprises		Conditions	
	Bull	Bear	Low	High	Low	High	High	Low	Positive	Negative	Good	Bad
Momentum												
Return	**5.53%**	**−2.91%**	**2.81%**	**−0.19%**	**2.35%**	**−1.38%**	1.89%	0.10%	1.46%	1.04%	1.73%	−1.51%
Volatility	**4.37%**	**5.87%**	**3.16%**	**7.08%**	3.35%	3.42%	5.39%	5.32%	5.31%	4.80%	**4.59%**	**8.62%**
RTR	**1.39**	**−0.35**	**0.88**	**0.16**	**0.81**	**−0.35**	0.56	0.29	0.62	0.35	0.59	0.02
Investment												
Return	**4.88%**	**−2.50%**	**2.61%**	**−0.22%**	**2.40%**	**−1.40%**	1.57%	0.20%	1.39%	0.86%	1.57%	−1.34%
Volatility	**3.96%**	**5.31%**	**2.91%**	**6.36%**	3.29%	3.25%	4.85%	4.80%	4.62%	4.66%	**4.07%**	**8.35%**
RTR	**1.38**	**−0.36**	**0.88**	**0.14**	**0.79**	**−0.43**	0.53	0.30	0.63	0.30	0.59	0.00
Profitablitiy												
Return	**4.69%**	**−2.63%**	2.23%	−0.17%	**2.38%**	**−1.45%**	1.49%	0.07%	1.20%	0.75%	1.37%	−1.21%
Volatility	3.97%	5.26%	**2.89%**	**6.34%**	3.31%	3.29%	4.86%	4.81%	4.63%	4.61%	**4.07%**	**8.29%**
RTR	**1.30**	**−0.39**	**0.77**	**0.14**	**0.78**	**−0.43**	0.49	0.25	0.58	0.25	0.52	0.04

The tables reports the conditional performance of the factor-tilted indices (size, value, high momentum, low investment and high profitability) by the five regions (USA, Japan, UK, Developed Europe ex-UK, and Asia Pacific ex-Japan), for the period from May 2004 to June 2015, considering a rolling window of WS=500. Stated are average monthly return (*Return*), the average monthly volatility (*Volatility*) and the corresponding reward-to-risk ratio (*RTR*). For our analysis we consider six variables (Market Returns, Realized one month Volatility, Term spread, Credit Default Swap (CDS) spread, Economic Surprises Index and Economic Conditions Index) to proxy twelve market conditions which, based on economic judgment, are either favorable (bull market regime, low volatility regime, low credit risk regime, high interest rate regime, positive economic surprises, favorable economic conditions) or unfavorable (bear market regime, high volatility regime, high credit risk regime, low interest rate regime, negative economic surprises, unfavorable economic conditions) for equity markets. Refer to Sect. 1 for how we define the respective states of the world. The cells in bold represent the instances in which the hypothesis that the average return, average volatility and reward to risk ratio are equal in both states was rejected at a 99% confidence level

Positive/negative economic conditions Positive and negative economic conditions are differentiated by using the Merrill Lynch Economic Conditions Index, calculated for each region. The index is a composite coincident indicator of real economic activity in the respective area. For Asia ex-Japan, the index is calculated as a GDP-weighted average of the individual countries' Purchasing Manager Index (PMI) for China, India, Korea, Taiwan, Hong-Kong and Singapore. For the rest of the regions, the methodology used, mirrors the one currently employed by the Philadelphia Fed to track US real business conditions. Its underlying economic indicators (i.e., weekly jobless claims, monthly payroll employment, industrial production, personal income less transfer payments, manufacturing and trade sales, quarterly Gross Domestic Product) blends high and low frequency information and stock and flow data, allowing continuous assessment of the state of the economy. A positive Economic Conditions index value means better than average economic conditions and a negative Economic Conditions index value means worse than average economic conditions.

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